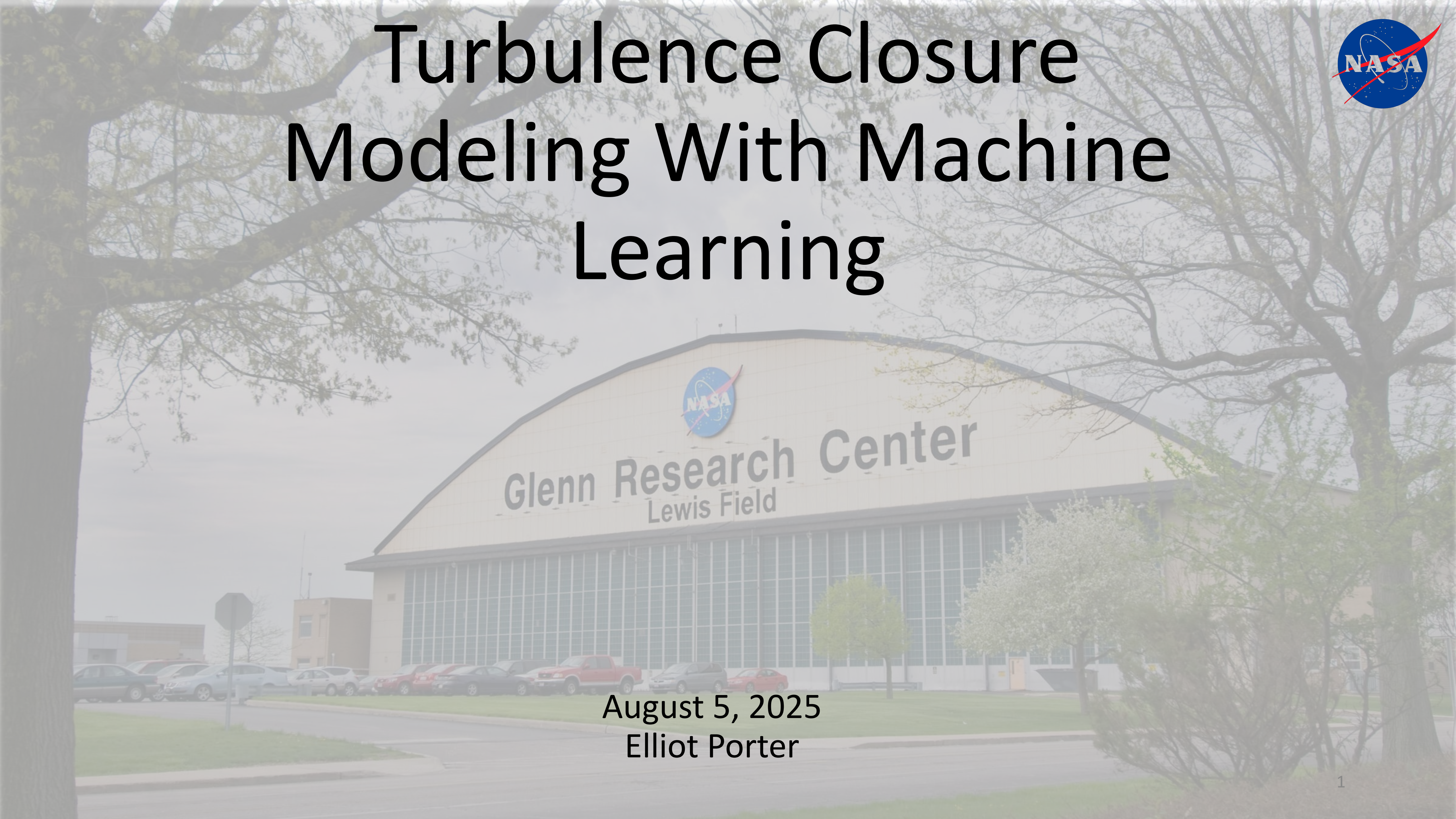
Turbulence Closure Modeling With Machine Learning

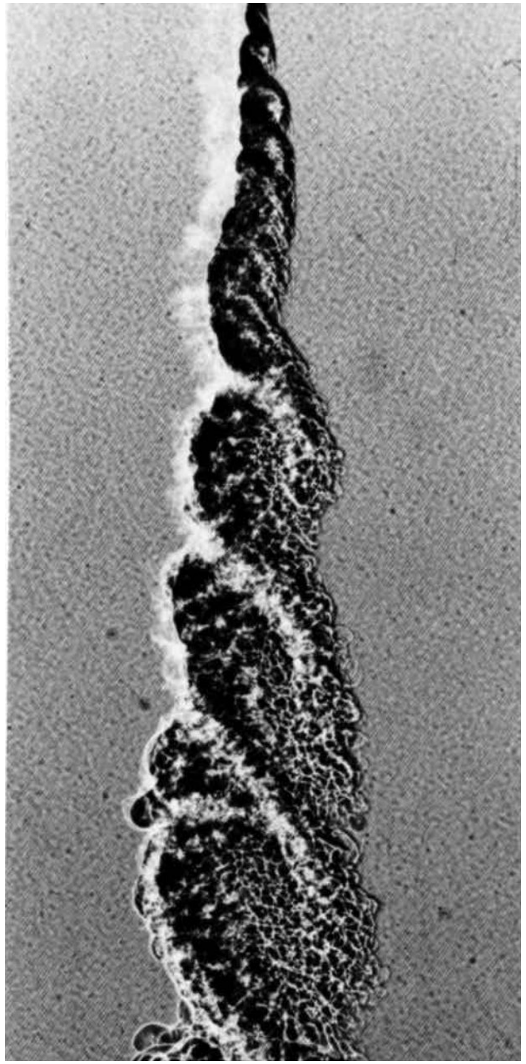


August 5, 2025

Elliot Porter

Agenda

Topic
Motivation
Problem Overview and Datasets
Machine Learning Overview
Feature Engineering and Assumptions
Trials and Results
Next Steps



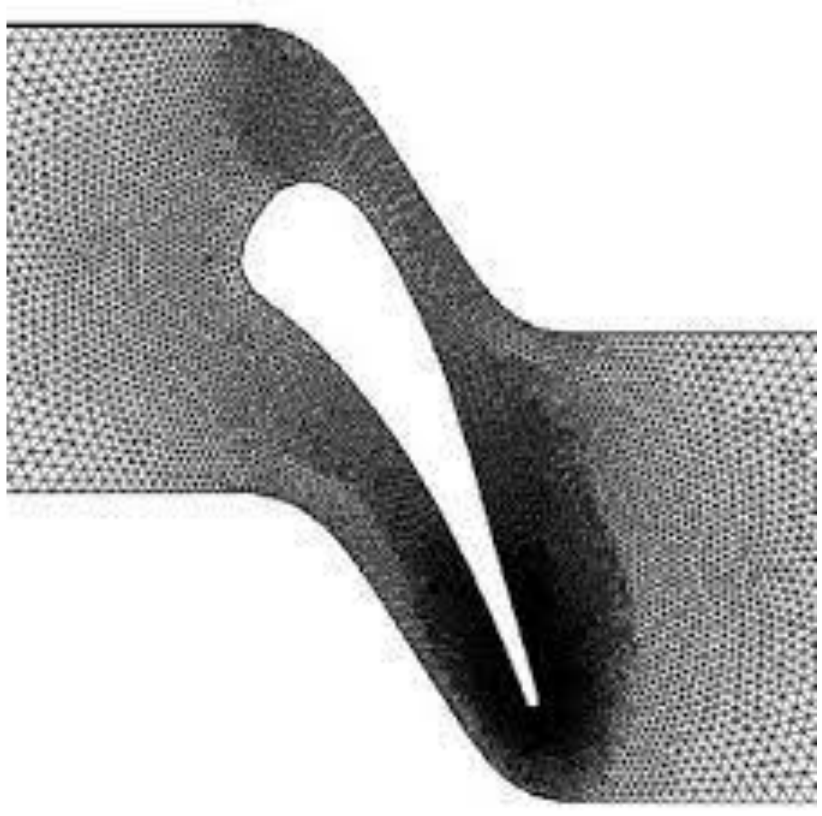
Van Dyke: Album of Fluid Motion p.102

Motivation

In Turbomachinery, the focal point is the trailing edge of the turbine blade.

RANS turbulence models fail to close on Reynolds stress anisotropy.

RANS solvers struggle to produce indicative anisotropy for shear mixing layers.



RANS Shortcomings

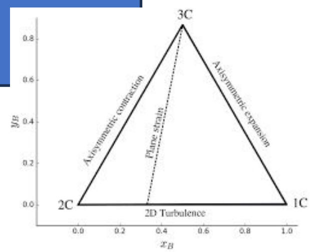
RANS solvers apply the Boussinesq approximation, which ties anisotropy to mean strain rate.

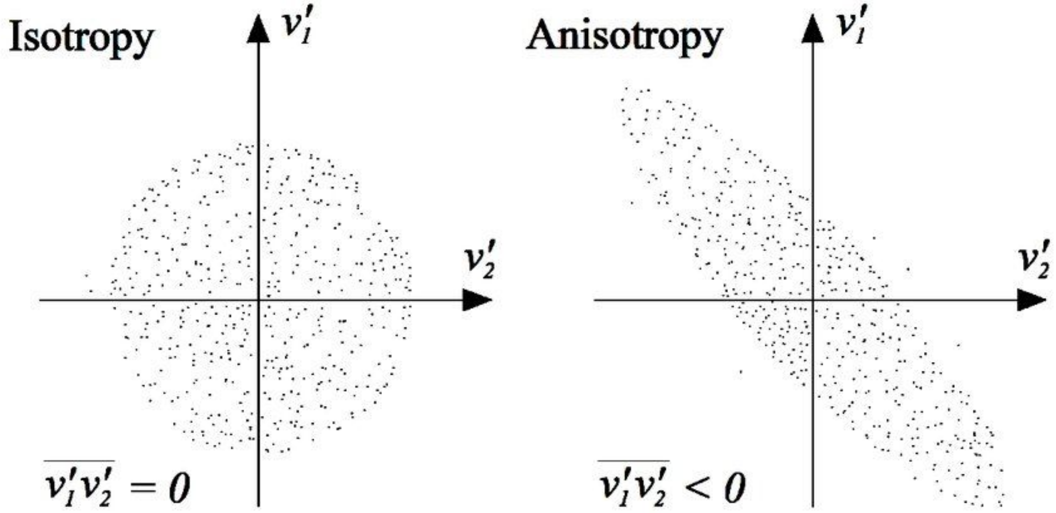
$$\frac{\partial}{\partial t}(\bar{\rho}\,\tilde{u}_i) + \frac{\partial}{\partial x_j}(\bar{\rho}\,\tilde{u}_i\,\tilde{u}_j) = -\frac{\partial \bar{p}}{\partial x_i} + \frac{\partial \bar{\tau}_{ij}}{\partial x_j} - \frac{\partial \overline{\rho u_i'' u_j''}}{\partial x_j}$$

Boussinesq approximation:

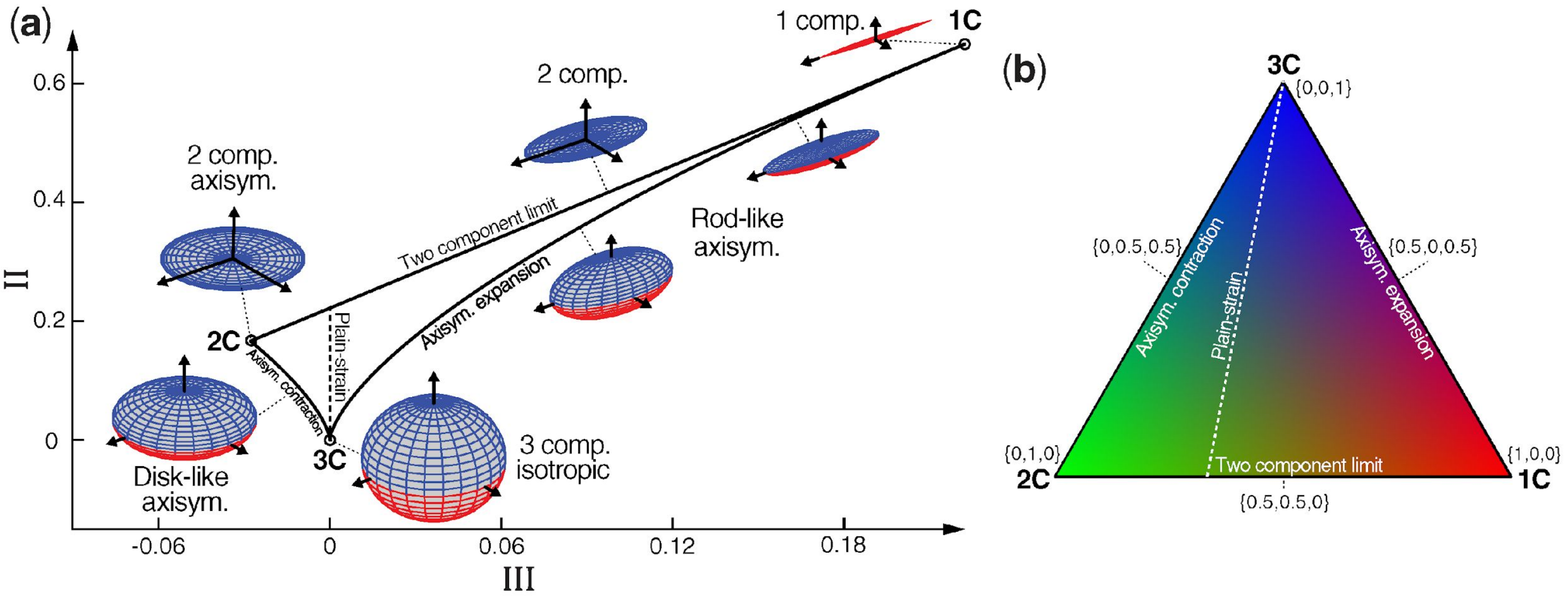
$$-\overline{\rho u_i'' u_j''} = \bar{\rho}\left(2\nu_t \tilde{S}_{ij} - \frac{2}{3}k\delta_{ij}\right)$$

The Boussinesq hypothesis confines RANS to the 'plane-of-strain' of the Lumley Triangle

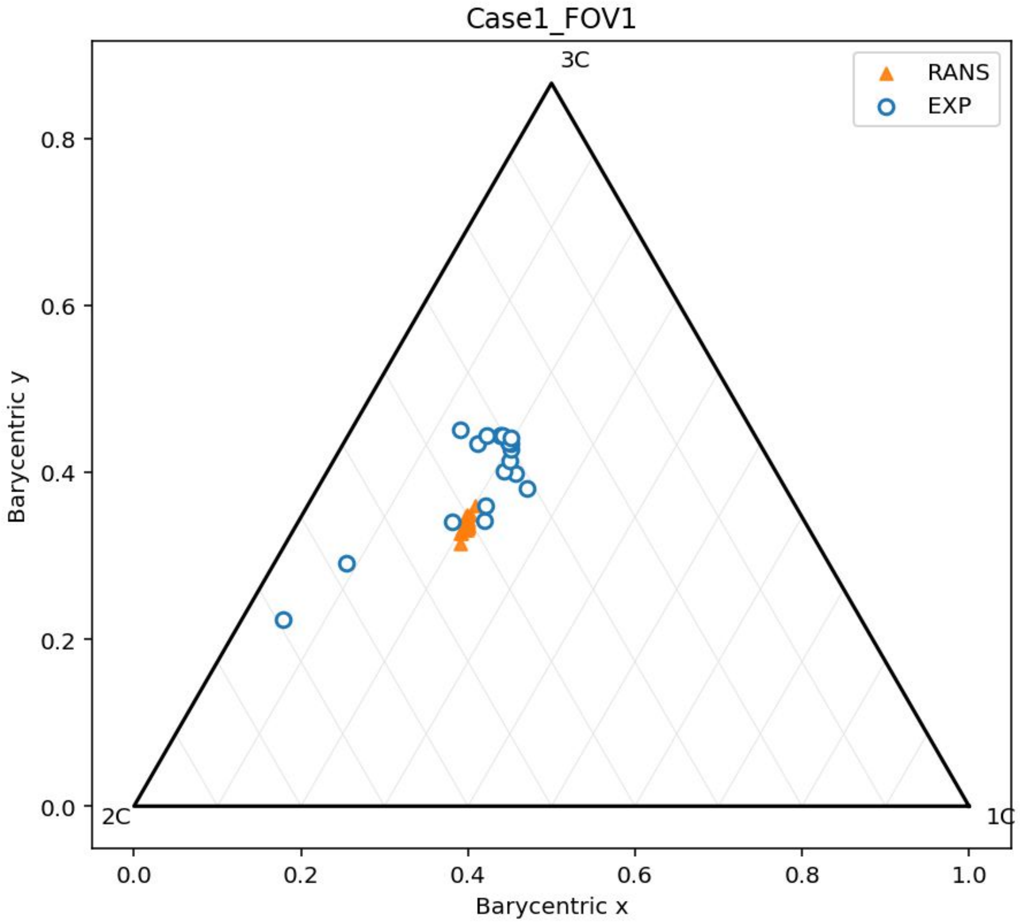


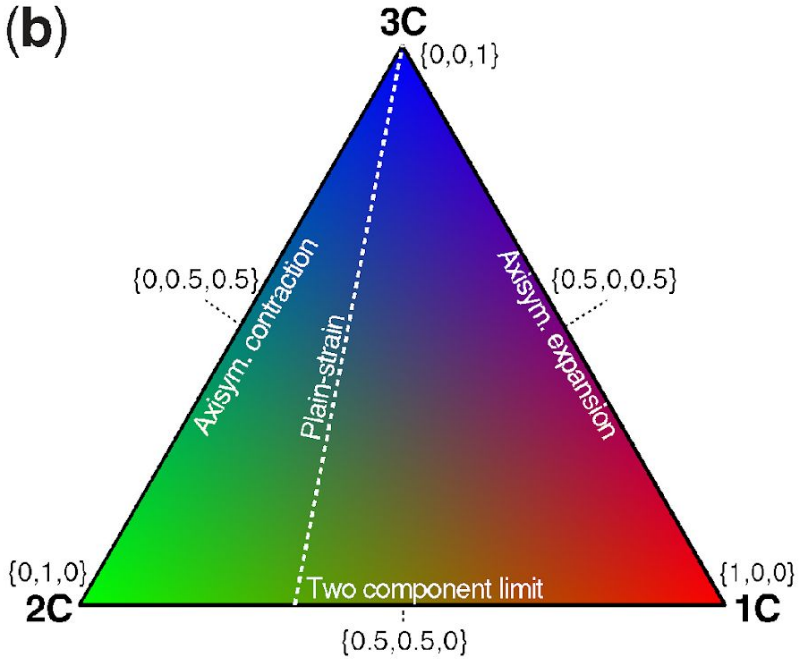


The Lumley Triangle



The Lumley Triangle





Problem Overview

We want to create a Machine Learning Model that:

- Inputs RANS data (of model that uses eddy viscosity)
- Outputs anisotropy tensor
- Retains turbulent kinetic energy
- Learns the physical constraints of anisotropy

ML Brief

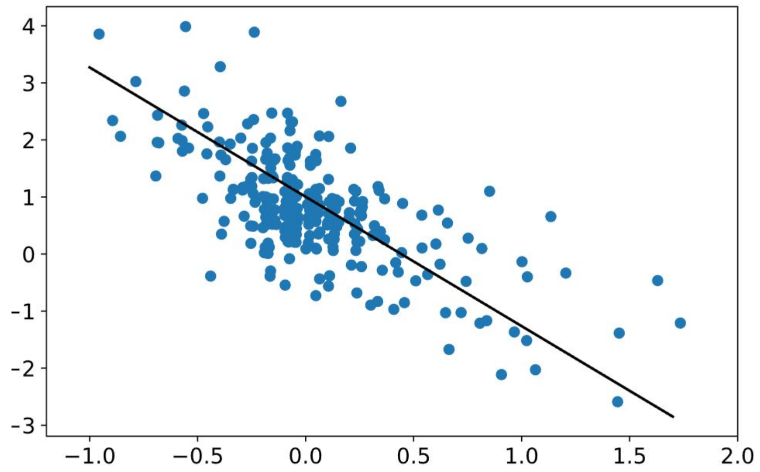
- A ML model learns nonlinear mappings between inputs and outputs

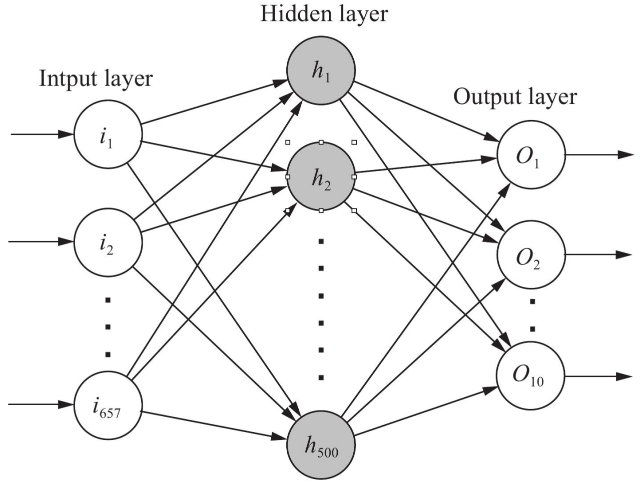
- Think of Linear Regression if X and Y were both multidimensional

- A model will “learn” over the course of training on datasets

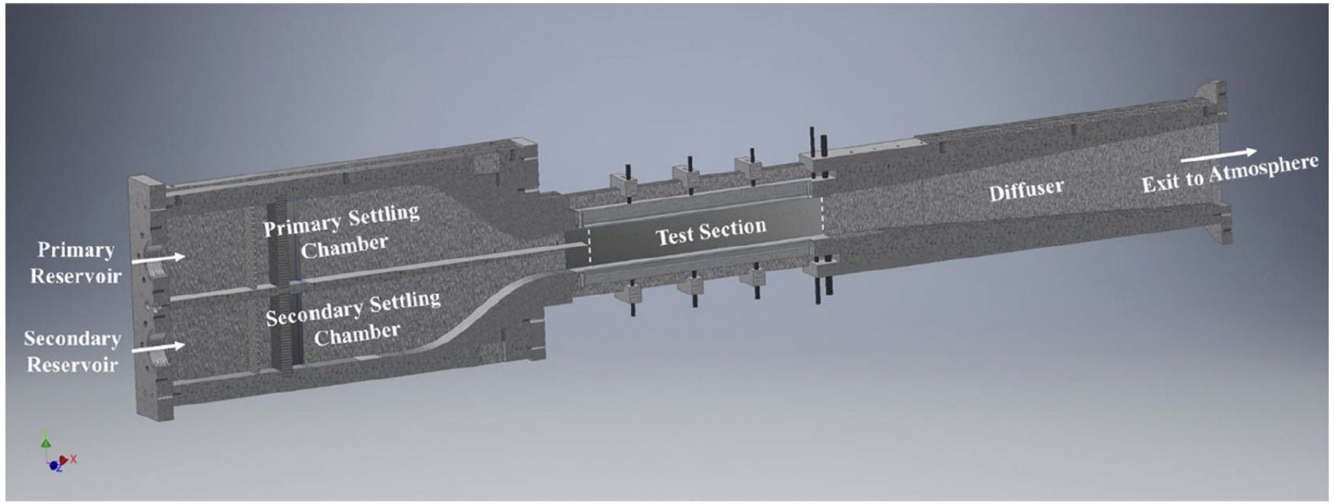
- Each iteration the model will gauge how wrong it is and update the network

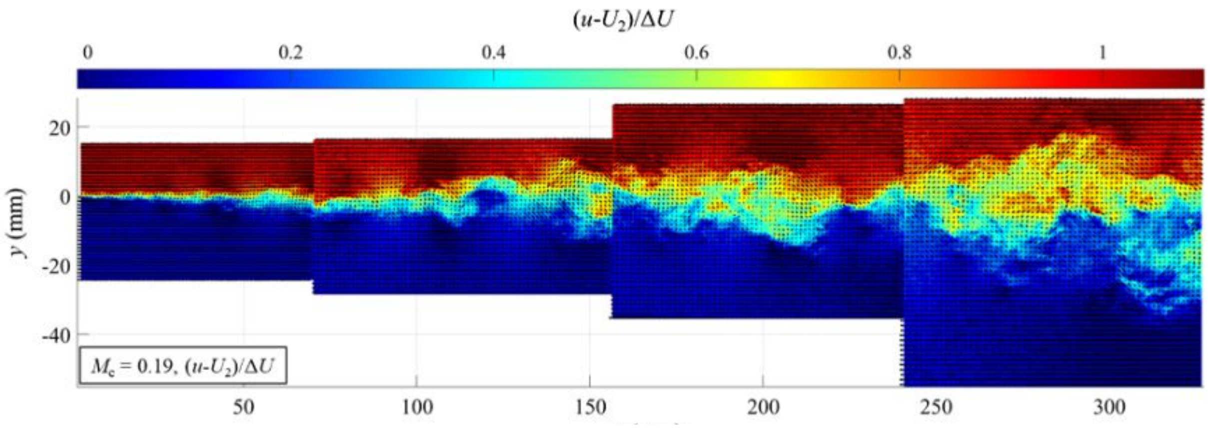
loss functions are how we define how wrong the model is





Dataset: NASA UIUC Compressible Mixing Layers





Input Parameters	Case 1	Case 2	Case 3
M_1	0.463 ± 0.012	1.003 ± 0.021	1.571 ± 0.025
M_2	0.089 ± 0.009	0.189 ± 0.009	0.285 ± 0.014
P_{01} (kPa)	109.32 ± 0.05	151.84 ± 0.05	270.41 ± 0.05
P_{02} (kPa)	94.56 ± 0.05	82.47 ± 0.05	71.47 ± 0.05
P_1 (kPa)	93.94 ± 0.10	80.37 ± 0.11	62.02 ± 0.10
P_2 (kPa)	93.92 ± 0.10	80.59 ± 0.11	66.29 ± 0.10
T_{01} (K)	296.10 ± 0.50	294.60 ± 0.50	284.59 ± 0.50
T_{02} (K)	292.48 ± 0.50	293.58 ± 0.50	295.57 ± 0.50
T_1 (K)	283.95 ± 0.84	245.25 ± 2.04	190.50 ± 2.68
T_2 (K)	292.02 ± 0.51	291.49 ± 0.54	290.85 ± 0.69
U_1 (m/s)	156.25 ± 4.3	314.89 ± 6.32	434.76 ± 6.08
U_2 (m/s)	30.35 ± 3.0	64.75 ± 2.96	97.28 ± 4.87
$\boldsymbol{r = U_2/U_1}$	**0.194 ± 0.020**	**0.206 ± 0.010**	**0.224 ± 0.012**
$\boldsymbol{s = \rho_2/\rho_1}$	**0.972 ± 0.004**	**0.844 ± 0.007**	**0.700 ± 0.010**
$\boldsymbol{M_c}$	**0.185 ± 0.008**	**0.381 ± 0.011**	**0.546 ± 0.013**

Data Processing and Feature Engineering Overview

Experimental Datasets

- Run RANS on each experimental setup
- Derive Reynolds Stress Anisotropy
- Perform Non dimensional Scaling
- Normalize to be interpretable by the Model

RANS Datasets

- Derive terms that describe the flow
- Make model input feature sets
- Perform Non dimensional scaling
- Normalize for interpretability

Feature Engineering Assumptions

Ideal Gas

$$\rho = \frac{p}{R\,T}$$

Sutherland Viscosity

Viscosity is a function of Temperature

$$\mu(T) = \mu_0 \left(\frac{T}{T_0}\right)^{3/2} \left(\frac{T_0 + S}{T + S}\right)$$

Negligible density Fluctuations

$$\overline{\rho u_i' u_j'} \approx \overline{\rho} \cdot \overline{u_i' u_j'}$$

$$\overline{u_i'' u_j''} \approx \overline{u_i' u_j'} \quad (\text{Favre} \approx \text{Reynolds})$$

Justifications:

- Ideal at weak compressibility
- Smooth viscosity field, no nu_t
- Reduce the number of terms to close upon
- Low convective mach means pressure fluctuations are negligible.

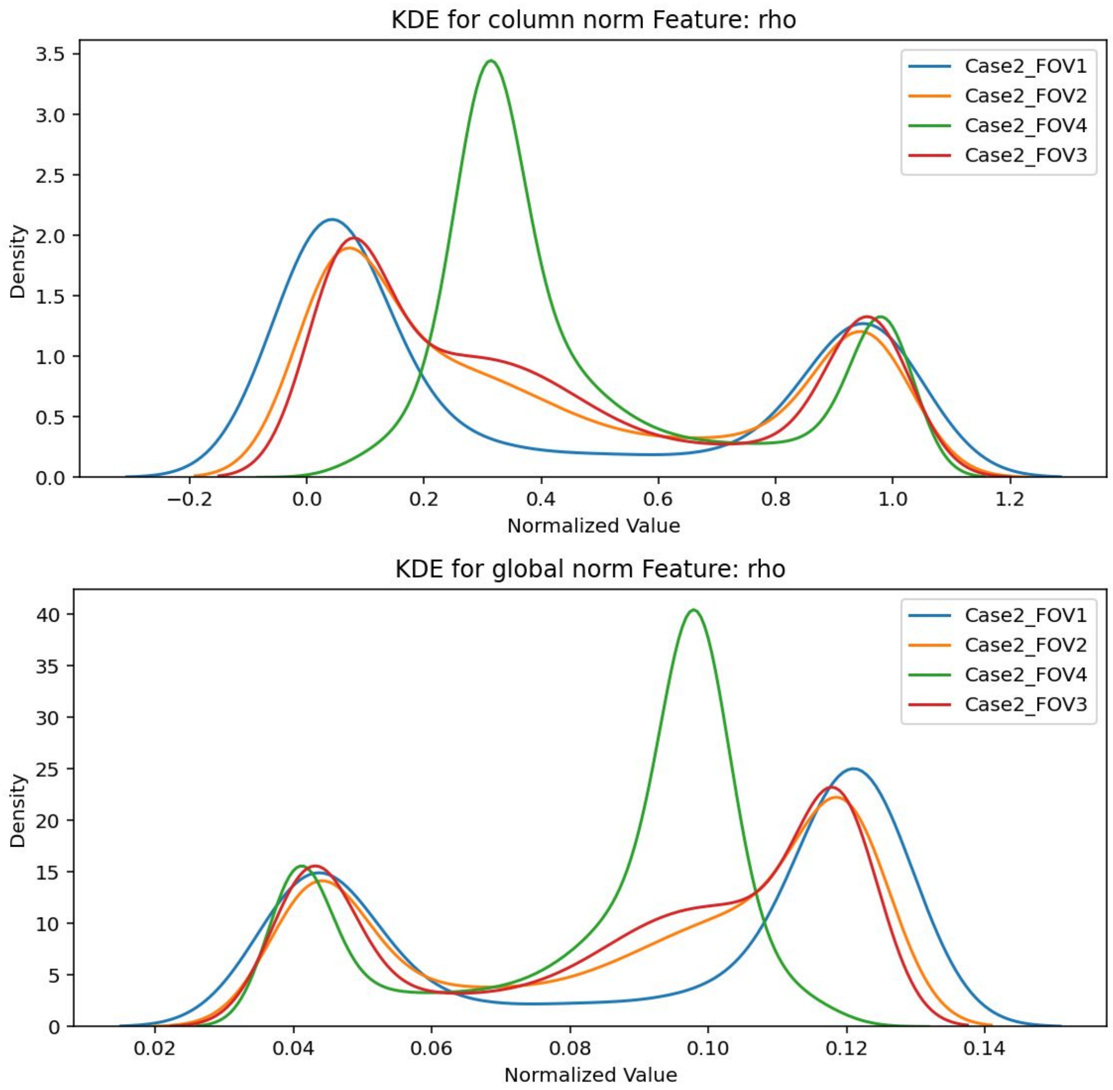
Feature Engineering – Normalization

Depending on your activation function for your network you need to normalize your data to its optimal working range.

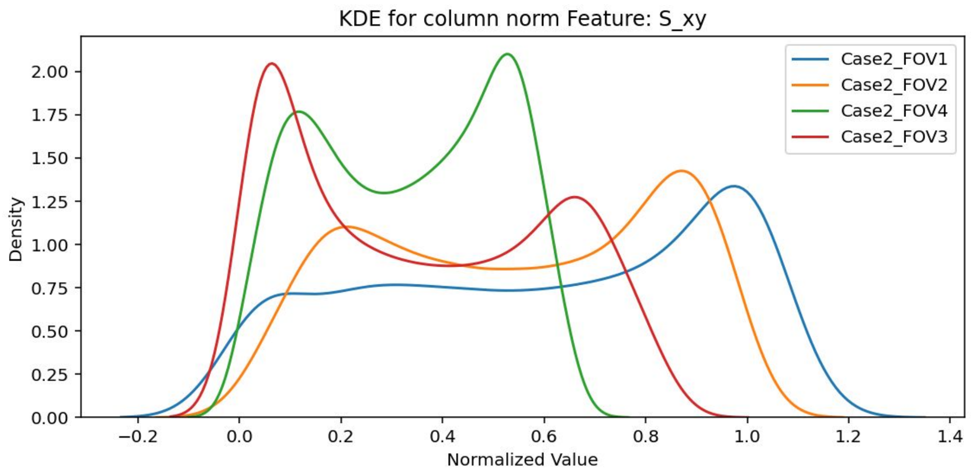
For our network we normalize 0-1 in accordance with leakyrelu activation.

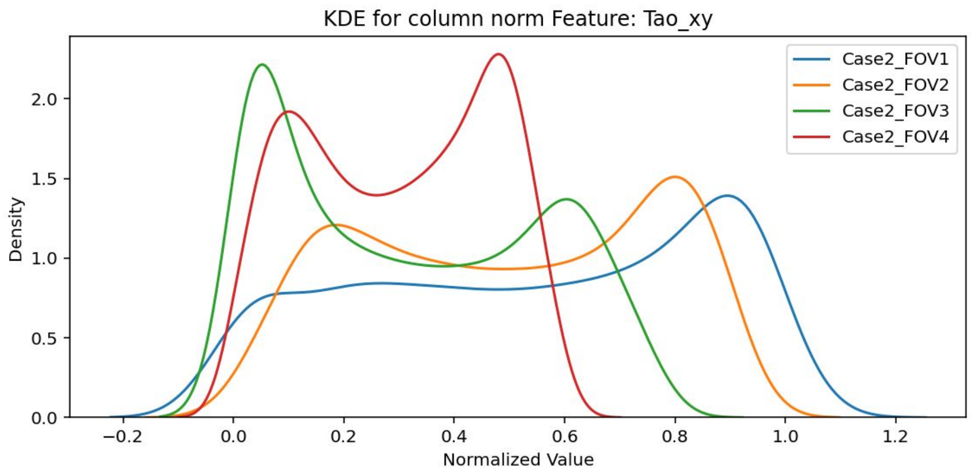
We use min-max scaling. Where we scale each column by its maximum* and minimum*.

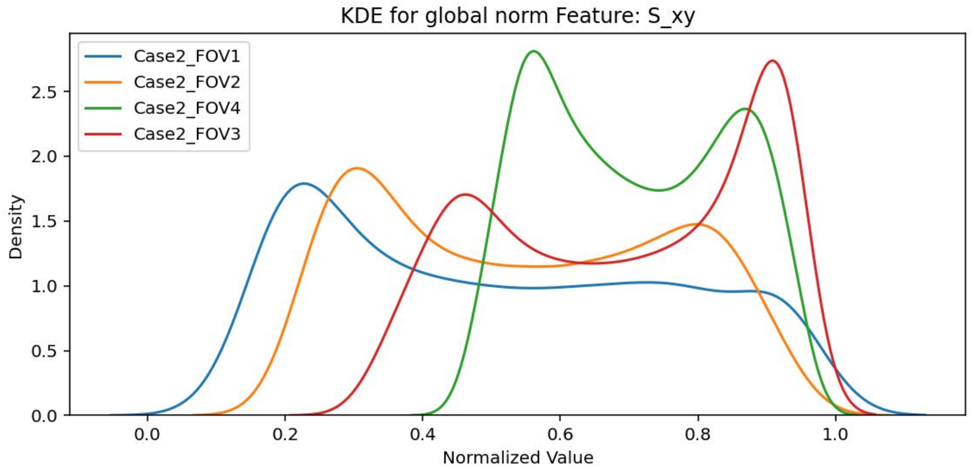
kurtosis clipping

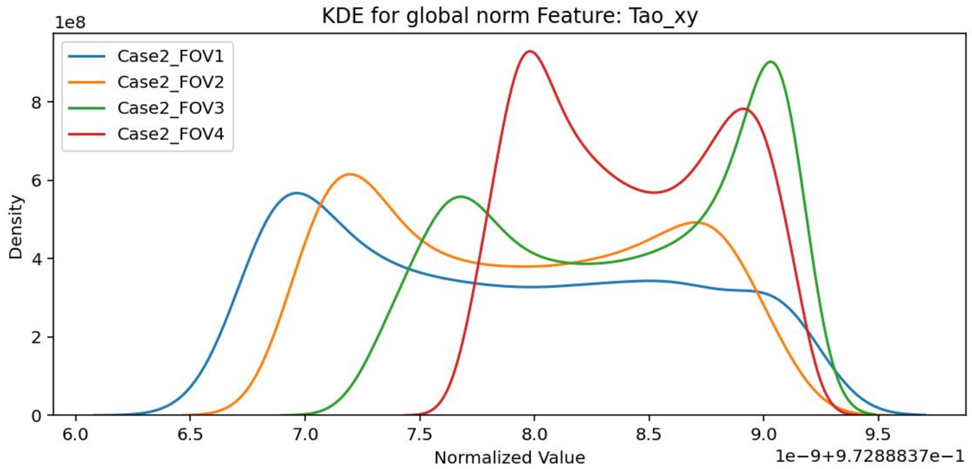


Feature Engineering – More KDE plots









Feature Engineering TLDR

#	Preprocess Steps	How and Why?
1	Take RANS data and feature engineered Terms from Favre Averaged RANS eq.	Derive features like rho, take Gradients. We want to close on the RST divergence term
2	Non Dimensionalize the features we pass in. Use boundary conditions to do so.	Puts everything in a generalizable form. Removes units and preserves scale.
3	Scale each feature across all cases to 0-1	Get global column min-max values. Then scale by such.
4	Make KDE plots for each feature with train and testing case distributions	Validate our data is different valid to be learned

Model Training and Loss Metrics

Our Primary Goal is for our model to accurately predict the Reynolds stress anisotropy tensor components.

$$a_{ij} = R_{ij} - \frac{2}{3}k\delta_{ij}$$

Loss is calculated from the L2 norm or MSE of the prediction

Turbulent Energy

$$k = \frac{1}{2}\overline{u_k'' u_k''}$$

Defined as half the trace of the RST

Eigen Invariants of a_{ij}:

$$I = \mathrm{tr}(a_{ij}) = 0$$

$$II = \frac{1}{2}a_{ij}a_{ji} = \frac{1}{2}a_{ji}a_{ij}$$

$$III = \frac{1}{3}a_{ij}a_{jk}a_{ki}$$

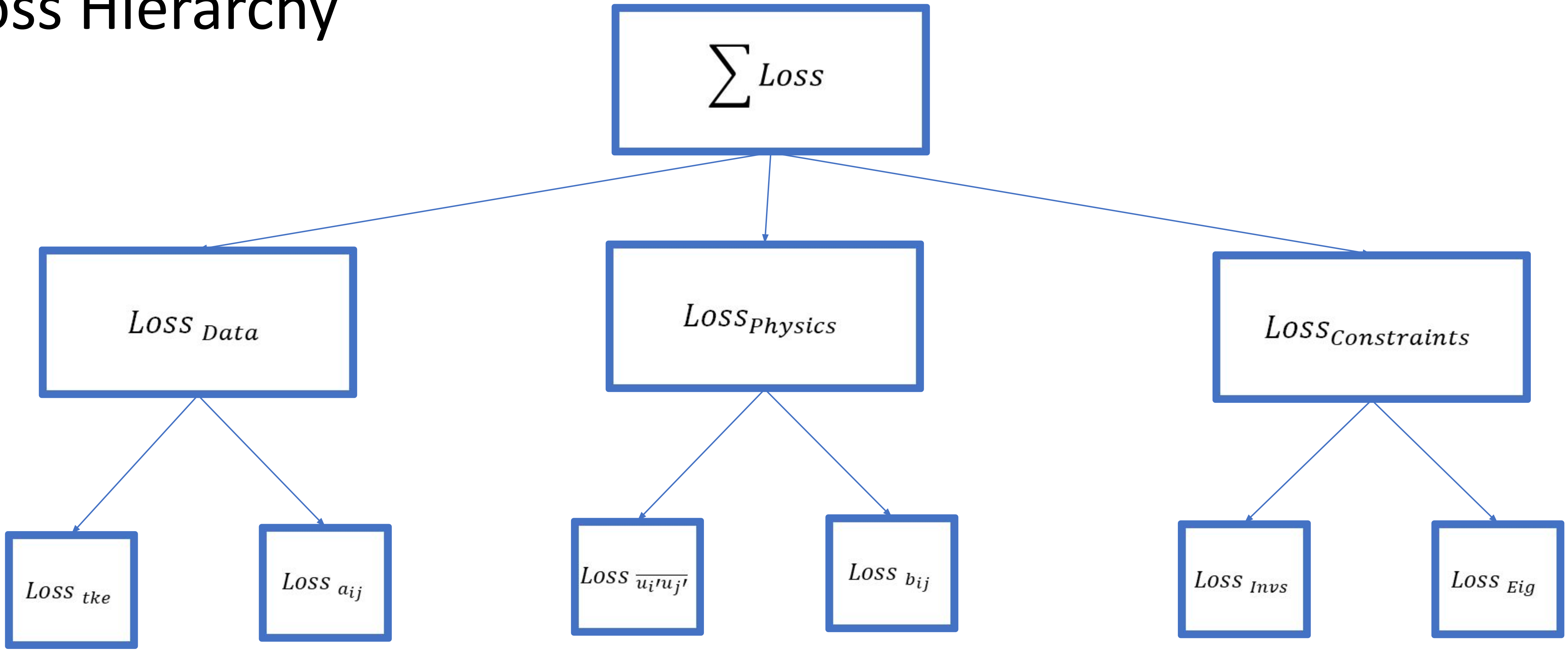
These are also equal to eigen values of aij

Normalized Anisotropy of RST

$$b_{ij} = \frac{a_{ij}}{2k} = \frac{R_{ij}}{2k} - \frac{1}{3}\delta_{ij}$$

Shape and Structure of turbulence.

Loss Hierarchy



Loss Between predicted and actual TKE

*Loss for each component of a_{ij} *

Loss for each component of R_{ij}, b_{ij}

Invariant Loss against LES invariants

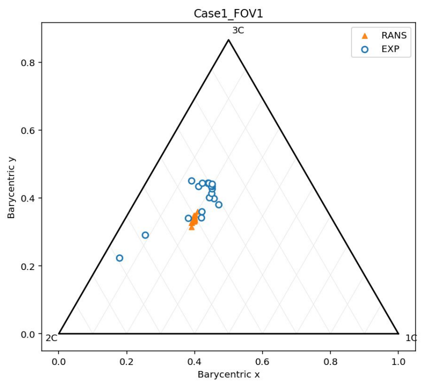
Eigen Value Loss against Pred Variants

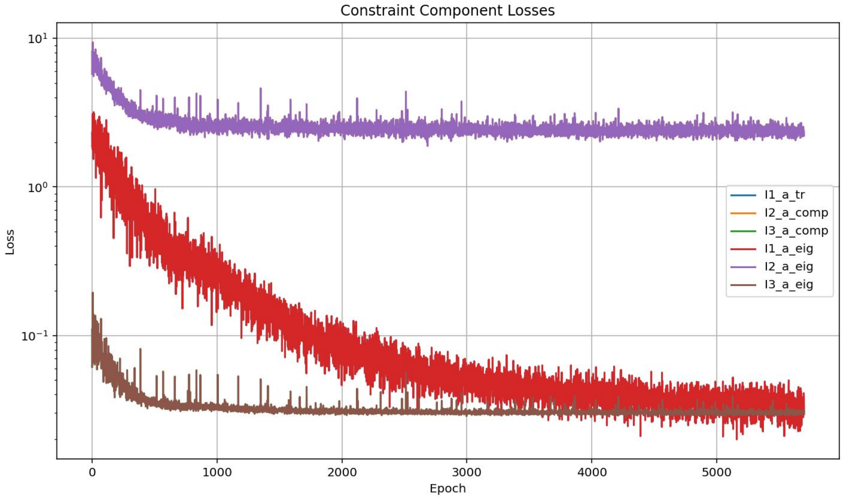
Model Training

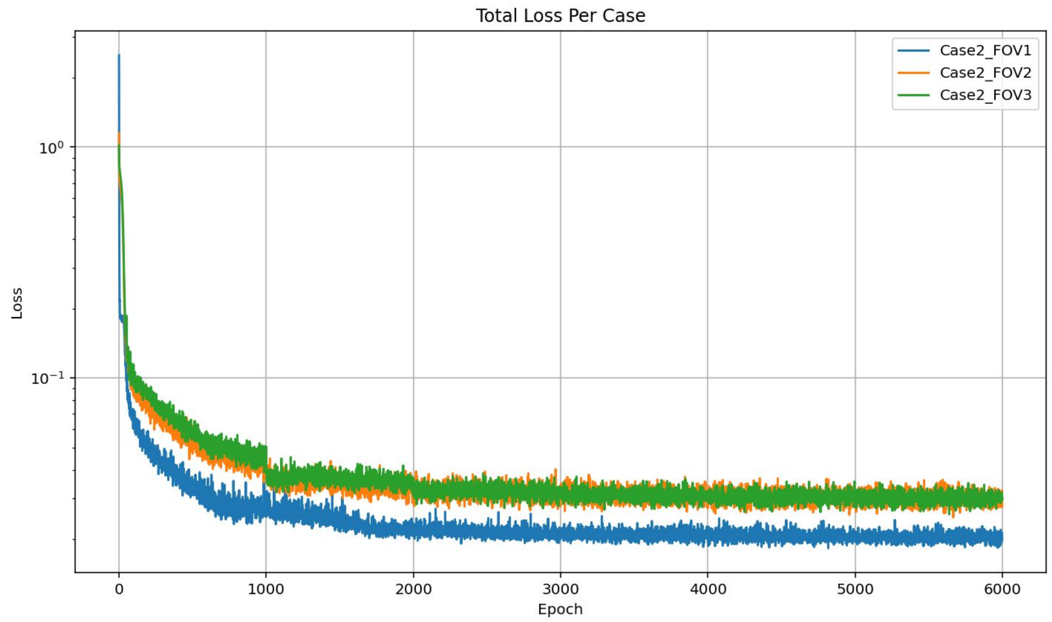
Trial Cases	Trial Type
Train: [Case1 FOV1, Case1 FOV2, Case 1 FOV 4] Test: [Case1 FOV3]	Interpolation
Train: [Case1 FOV1, Case1 FOV2, Case 1 FOV 3] Test: [Case1 FOV4]	Extrapolation
NB: Remember FOV index refers to downstream distance. Interpolating upstream is more difficult than downstream Extrapolating upstream is even more difficult than downstream.	

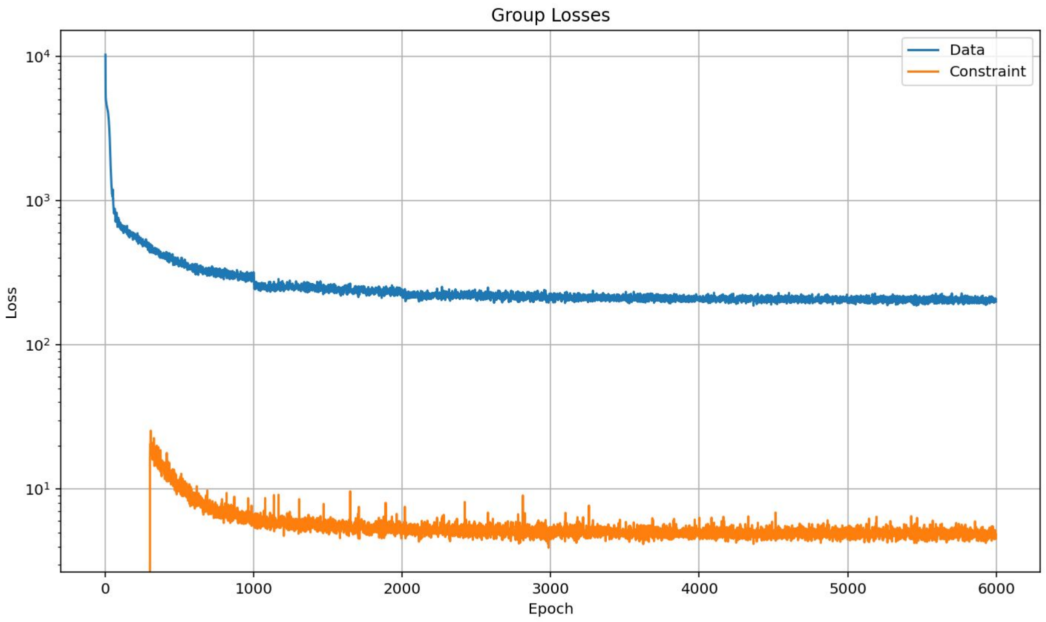
Results - Extrapolation

Valid Set	Terms
FS8	Base RANS Features (Ux, Uy, p dUi_dxj) Viscous stress tensor Rotation and Strain Tensors Advection Rotation Tensors

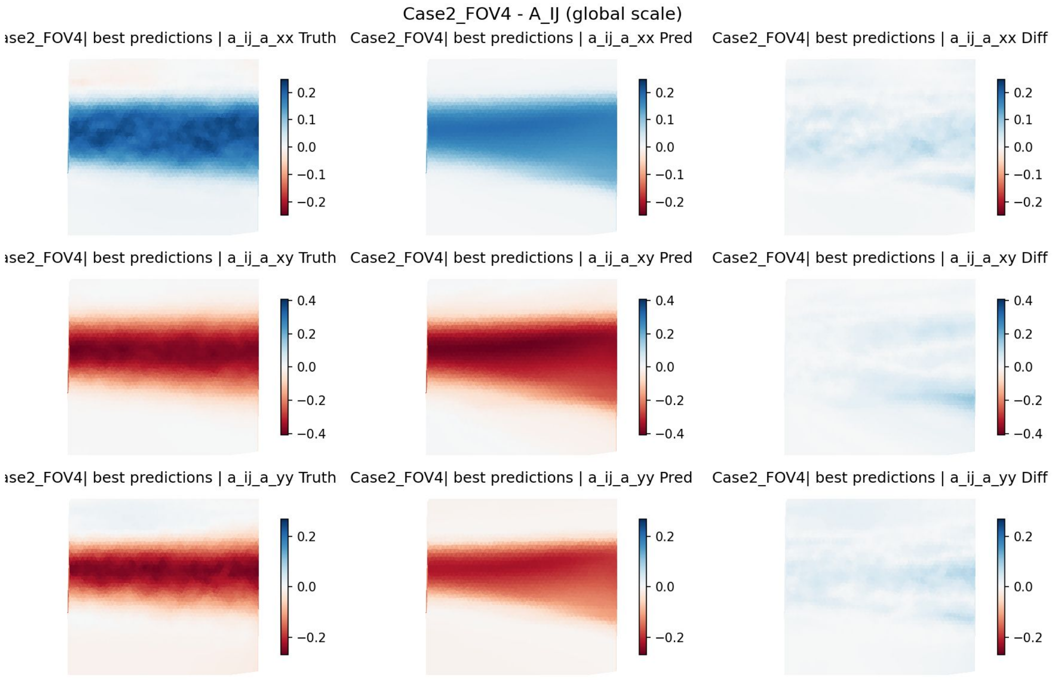


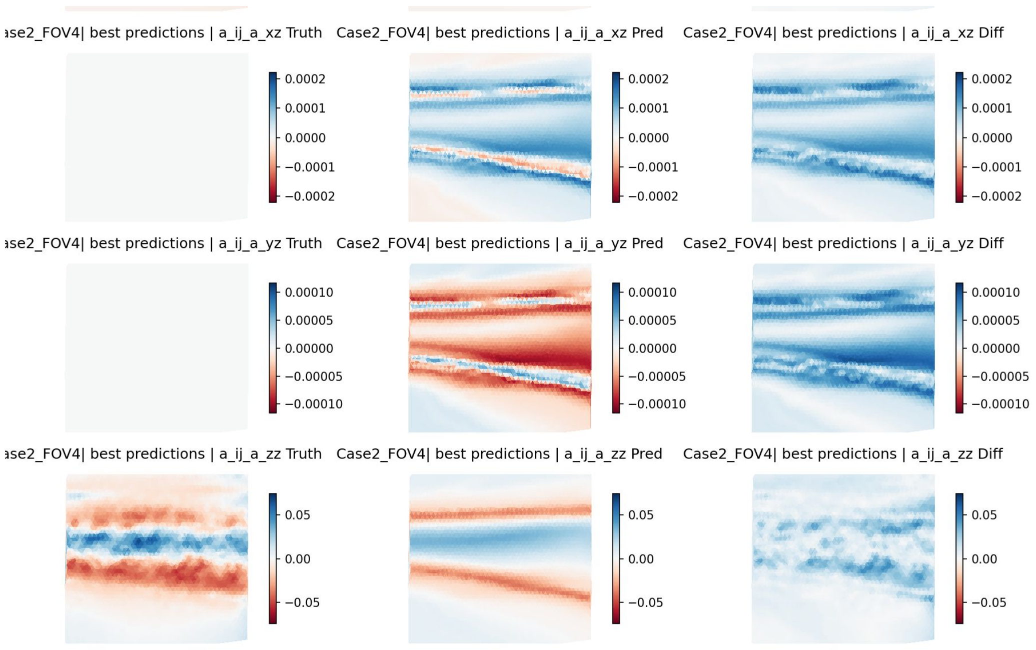




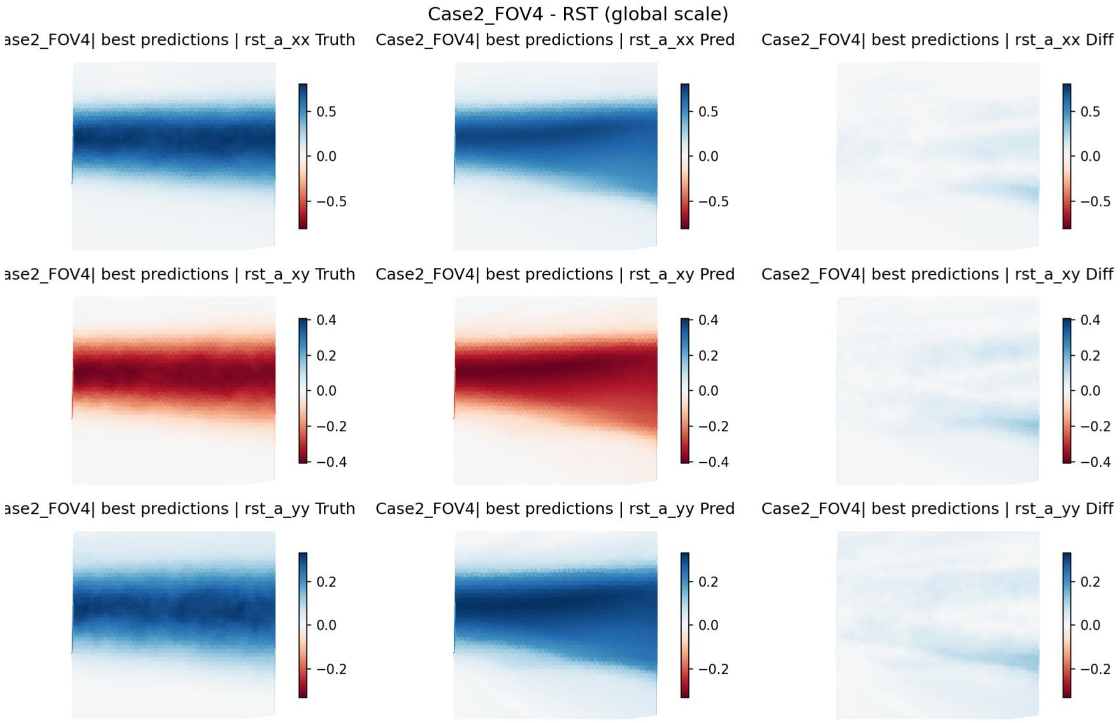


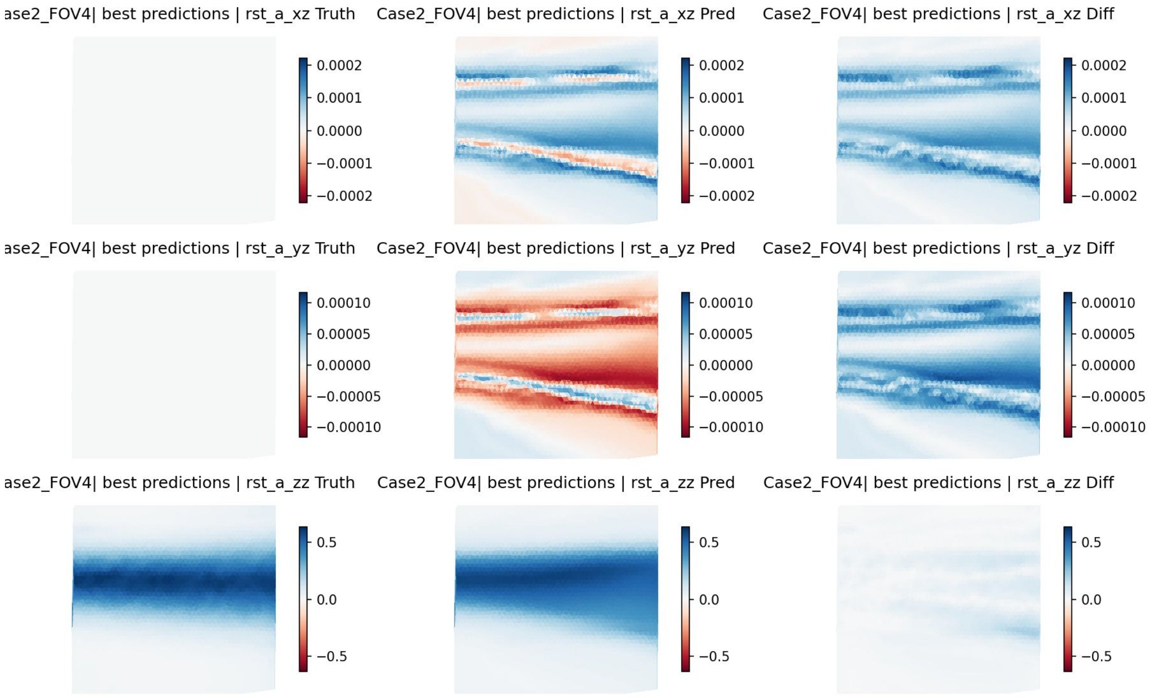
Results – Extrapolation TKE and Aij





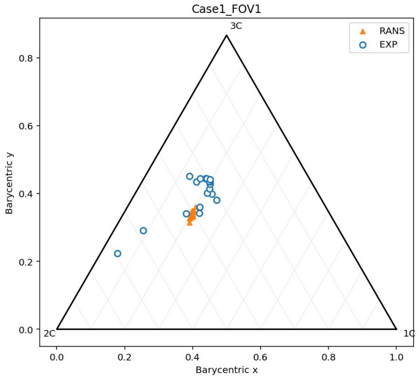
Results – Extrapolation RST

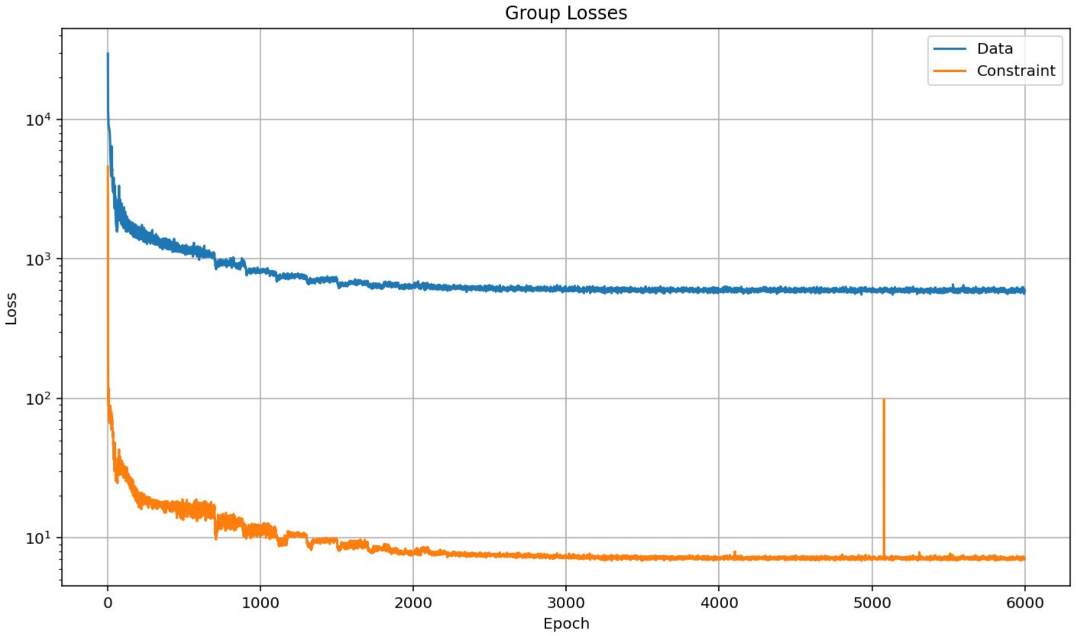


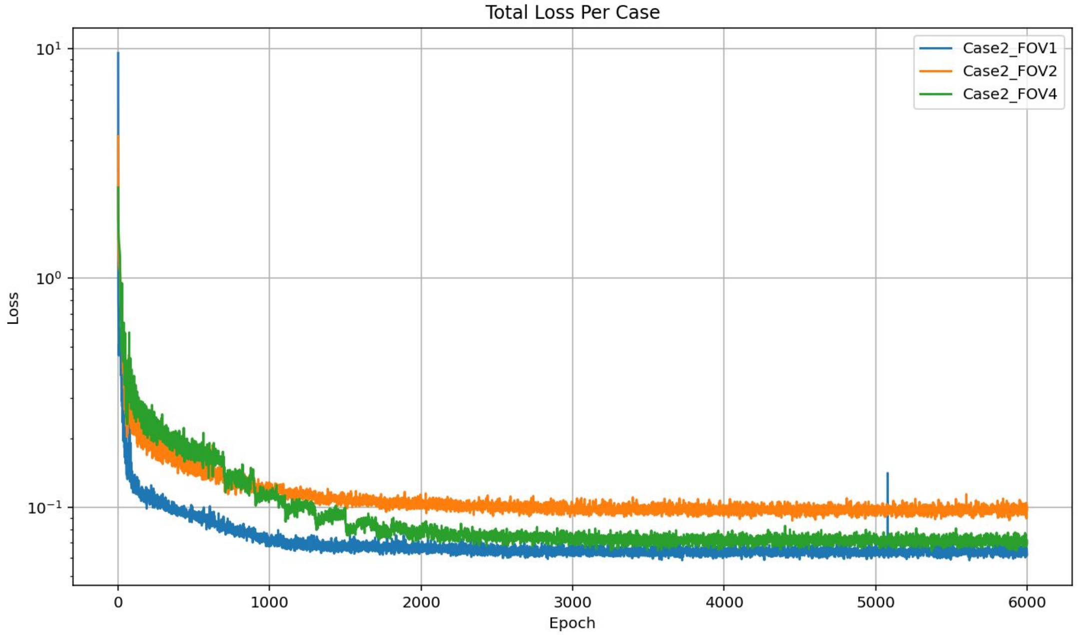


Results - Interpolation

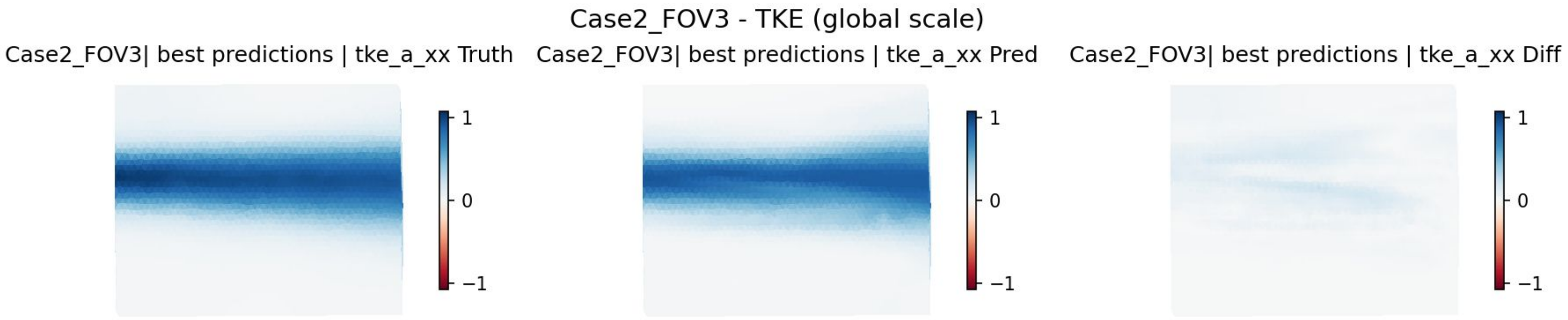
Valid Set	Terms
FS8	Base RANS Features (Ux, Uy, p dUi_dxj) Viscous stress tensor Rotation and Strain Tensors Advection Rotation Tensors

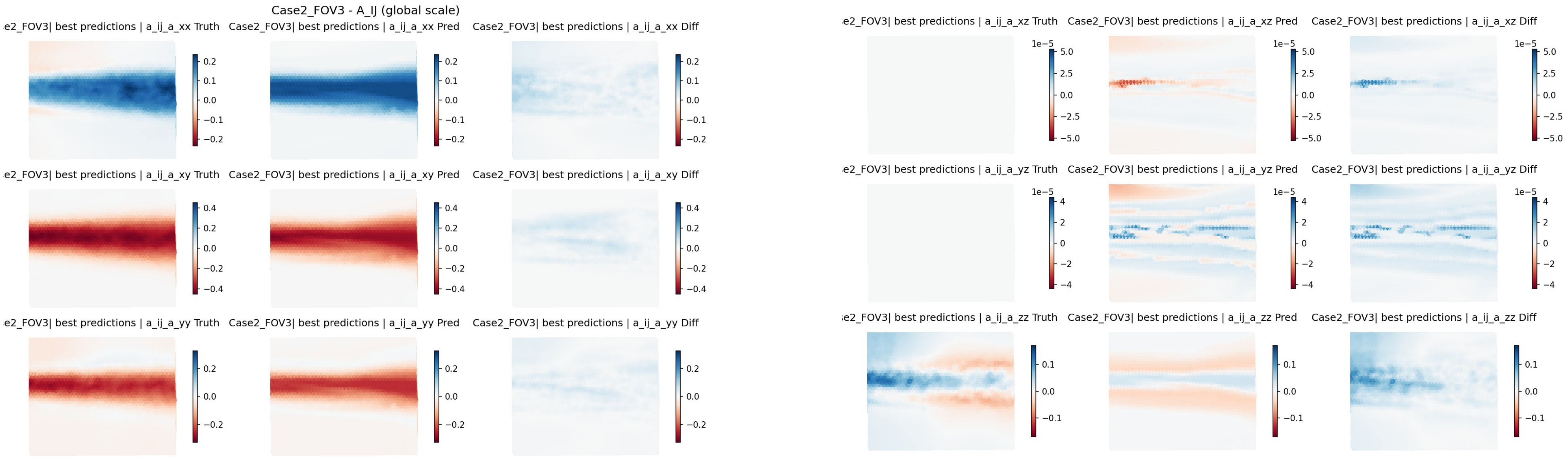




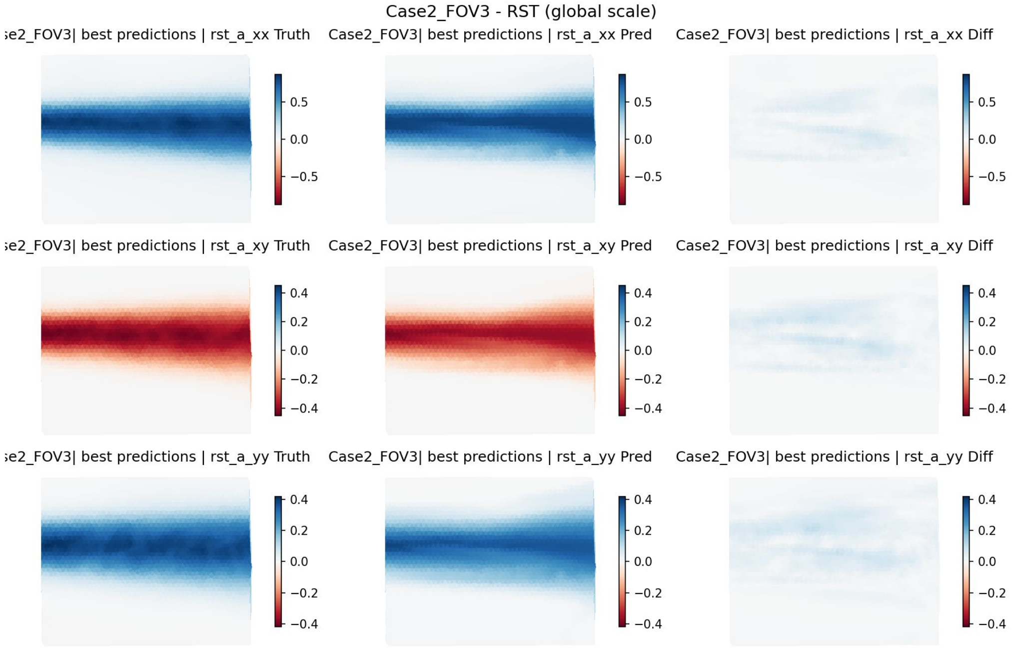


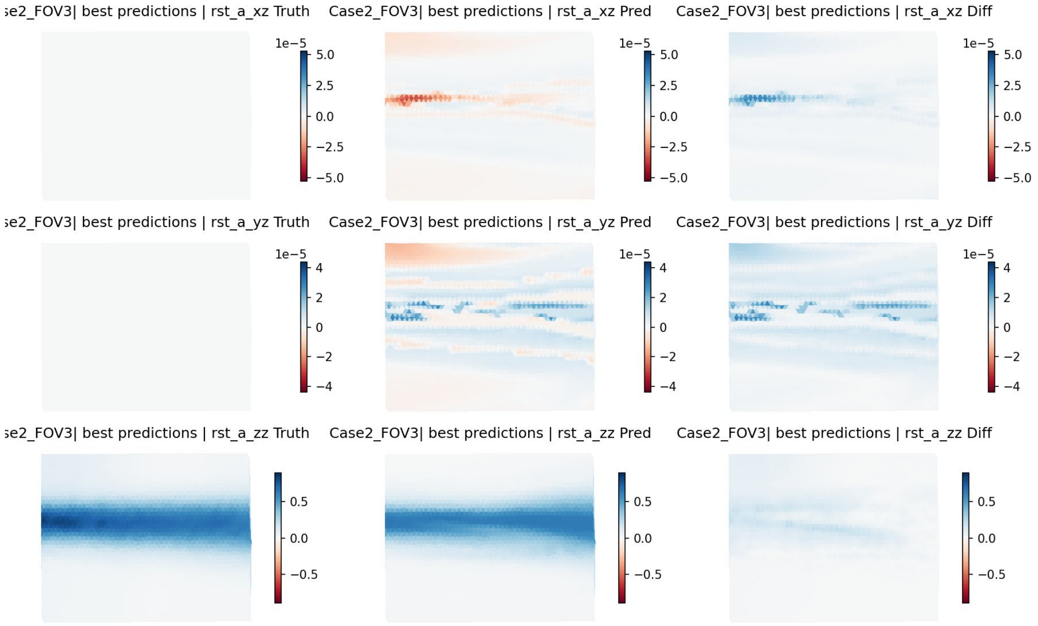
Results – Interpolation TKE and Aij





Results – Interpolation RST





Takeaways

1. We can recover Reynolds stress anisotropy from RANS data with accuracy
2. ML methodology used can be applied to many simulation experiment modeling
3. Formulating loss by groups helps model convergence

Next Steps

1. Implement Closure Metrics
   Validate inputs vs outputs for closure on FARANS equations
2. Reevaluate density fluctuations assumption
   alternative ways to model
3. Modeling FARANS energy with turbulent heat fluxes
   Bridging flow and HT crucial for generalizable modeling
4. Apply model to trailing edge of Turbine Blade
   To refine stresses present (validate w LES/DNS)

Questions?

August 5, 2025
Elliot Porter

References

1. Kim, Kevin U., Elliott, Gregory S., and Dutton, J. Craig, "Compressibility Effects on Large Structures and Entrainment Length Scales in Mixing Layers," **AIAA Journal**, Vol. 58, No. 12, 2020, pp. 5168-5182, https://doi.org/10.2514/1.J059433.

2. Dutton, J. Craig, Elliott, Gregory S., and Kim, Kevin U., "Compressible Mixing Layer Experiments for CFD Validation," AIAA-2019-2847, June 2019, https://doi.org/10.2514/6.2019-2847.

3. Dutton, J. C. and Elliott, G. S., "Final Report: Benchmark Experimental Measurements of Turbulent Compressible Mixing Layers for CFD Validation," Department of Aerospace Engineering, University of Illinois at Urbana-Champaign, NASA Collaborative Agreement NNX15AU94A, December 2020, PDF file.

4. Exp: Turbulent Compressible Mixing Layers (dataset)

Feature Engineering – Tensor Analysis

We take the Favre Equations, at steady conditions

We take the terms of these equations and calculate them with the RANS information that we do have.
The goal is to assemple the right combination of features such that the model can learn effectively.

This is where the physics is encoded.

[need to make slide prettier]

$$\frac{\partial}{\partial x_j}(\bar{\rho}\,\tilde{u}_i\,\tilde{u}_j) = -\frac{\partial \bar{p}}{\partial x_i} + \frac{\partial \bar{\tau}_{ij}}{\partial x_j} - \frac{\partial \overline{\rho u_i'' u_j''}}{\partial x_j}$$

Strain and Rotation Rates

$$S_{ij} = \frac{1}{2}\left(\frac{\partial \tilde{u}_i}{\partial x_j} + \frac{\partial \tilde{u}_j}{\partial x_i}\right) \qquad R_{ij} = \frac{1}{2}\left(\frac{\partial \tilde{u}_i}{\partial x_j} - \frac{\partial \tilde{u}_j}{\partial x_i}\right)$$

$$\bar{\tau}_{ij} = 2\mu\, S_{ij} - \frac{2}{3}\,\mu\,\delta_{ij}\,\frac{\partial \tilde{u}_k}{\partial x_k}$$

$$\bar{\rho}\,\tilde{u}_j\,\frac{\partial \tilde{u}_i}{\partial x_j} = \bar{\rho}\,\tilde{u}_j S_{ij} + \bar{\rho}\,\tilde{u}_j R_{ij}$$

Feature Engineering – Non-dimensional groups

The simplest way for a network to digest the information present is if it is stripped of dimensions.

Reference values were calculated from the boundary conditions of the given case of data we were working on. For inlets, mass flux averaging of the two inlets was applied for scalar field quantities.

Reference Scalars

$$\langle \phi \rangle_{\text{ref}} = \frac{\sum_i \rho_i U_i L_i \phi_i}{\sum_i \rho_i U_i L_i}$$

Non- Dimensional Groups

$$\phi^* = \frac{\phi}{\phi_{\text{ref}}}$$

$$\left(\frac{\partial u_i}{\partial x_j}\right)^* = \frac{L_{\text{ref}}}{U_{\text{ref}}} \frac{\partial u_i}{\partial x_j}$$

$$\frac{\mu u_\infty / L^2}{\rho_\infty u_\infty^2 / L} = \frac{\mu}{\rho_\infty u_\infty L} = \frac{1}{Re}$$

Favre equation

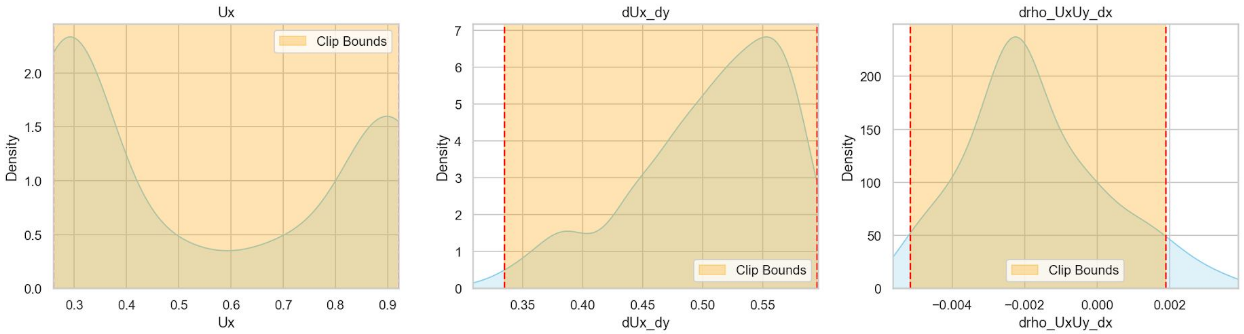
$$\frac{\partial(\bar{\rho}^* \tilde{u}_i^* \tilde{u}_j^*)}{\partial x_j^*} = -\frac{\partial \bar{p}^*}{\partial x_i^*} + \frac{1}{Re} \frac{\partial \bar{\tau}_{ij}^*}{\partial x_j^*} - \frac{\partial \overline{\rho^* u_i''^* u_j''^*}}{\partial x_j^*}$$

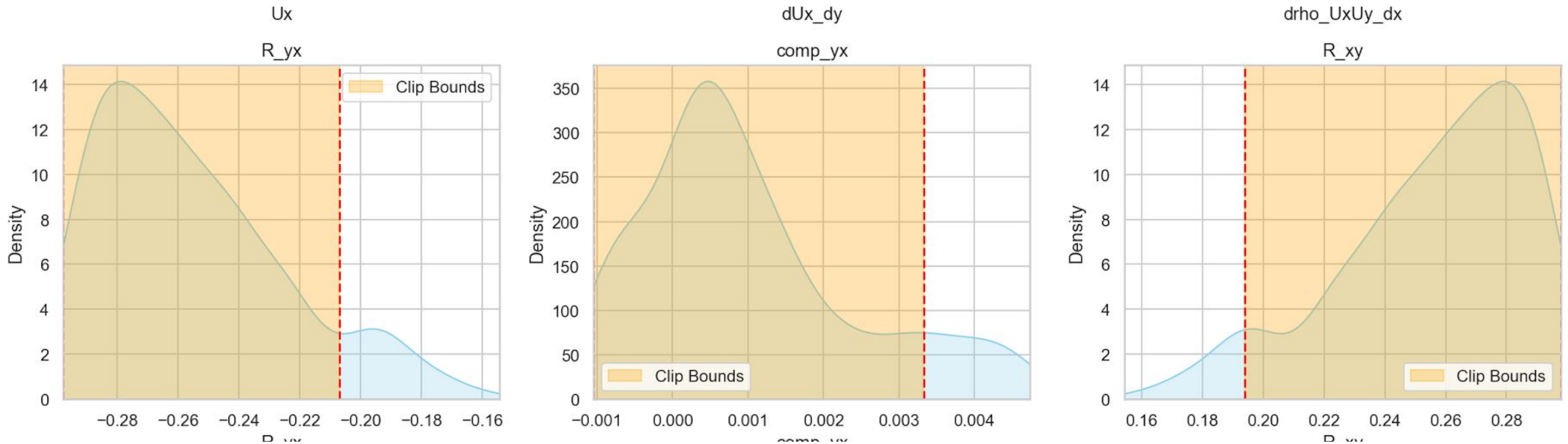
Template

Normalization Clipping

For features with large tails

- Clip the percentile from which to choose the maximum and minimum based on the tail direction





Adding Invariants of $\boldsymbol{a_{ij}}$ and $\boldsymbol{b_{ij}}$ as Constraints

Beyond just knowing the shape, magnitude and intensity of the turbulence, you need to make sure that the model learns to adhere to physical constraints.

While K (TKE) is technically the first invariant of the RST, in turbulence modeling the invariants of $\boldsymbol{a_{ij}}$ and $\boldsymbol{b_{ij}}$ provide more meaningful insight into turbulent structure:

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Adding Invariants of $\boldsymbol{a_{ij}}$ and $\boldsymbol{b_{ij}}$ as Constraints

<u>Invariants of a_{ij}:</u>

$$I = \mathrm{tr}(a_{ij}) = 0$$

$$II = \frac{1}{2} a_{ij} a_{ji} = \frac{1}{2} a_{ji} a_{ij}$$

$$III = \frac{1}{3} a_{ij} a_{jk} a_{ki}$$

<u>Invariants of b_{ij}:</u>

$$I = \mathrm{tr}(b_{ij}) = 0$$

$$II = b_{ij} b_{ji} = b_{ji} b_{ij}$$

$$III = b_{ij} b_{jk} b_{ki}$$

Rationale

***I* –** Recall that $\boldsymbol{a_{ij}}$ and $\boldsymbol{b_{ij}}$ are devioric components of the full RST. By definition they have zero trace.

***II* –** magnitude of anisotropy or normalized anisotropy

***III* –** Shape of the anisotropy and enables realizability constraints

Loss Weighting

We want to apply a weighting system to losses such that:

- Each Component of R_{ij}, a_{ij}, b_{ij} contribute equally to its group loss (Data, Phys, Const) regardless of Magnitude (applies to const and other phys loss components too)
- Make Each Case that we are training on contribute equally to loss
- Some loss groups more important than others. This can be static or a function of time (i.e. $\frac{cur\ Epoch}{Total\ Epochs}$)

How

$$w_{ij} = \frac{\max_{k,l}(\mu_{kl})}{\mu_{ij} + \varepsilon} \quad \text{where} \quad \mu_{ij} = \frac{1}{N}\sum_{n=1}^{N}\left|T_{ij}^{(n)}\right|$$

*Of tensor T_{ij} with N total points *

$$w_c = \frac{\max(\mu_k)}{\mu_k^{(c)} + \varepsilon}$$

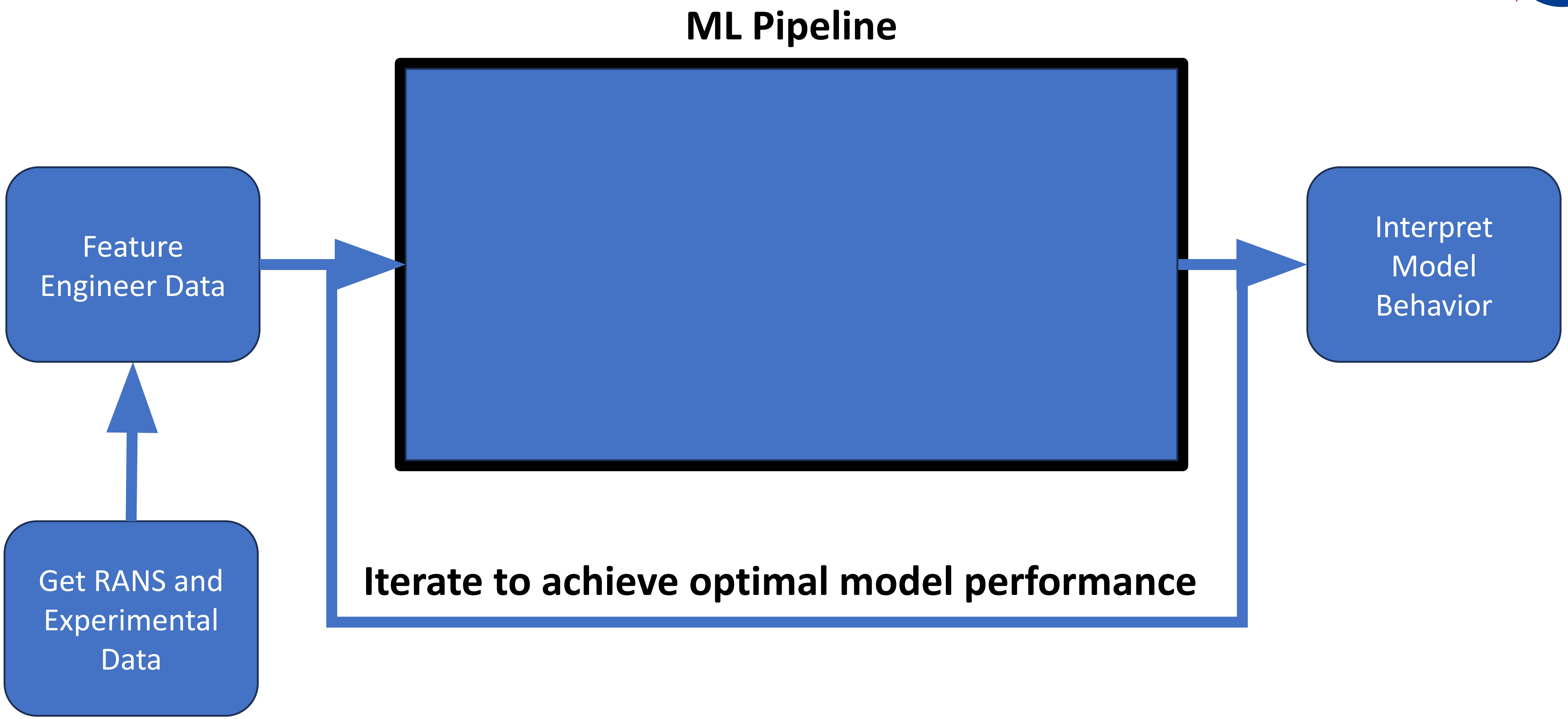
Scaling relative to TKE per N points

$$\sum Loss = \alpha_{Data}L_{Data} + \alpha_{phys}L_{phys} + \alpha_{const}L_{const}$$

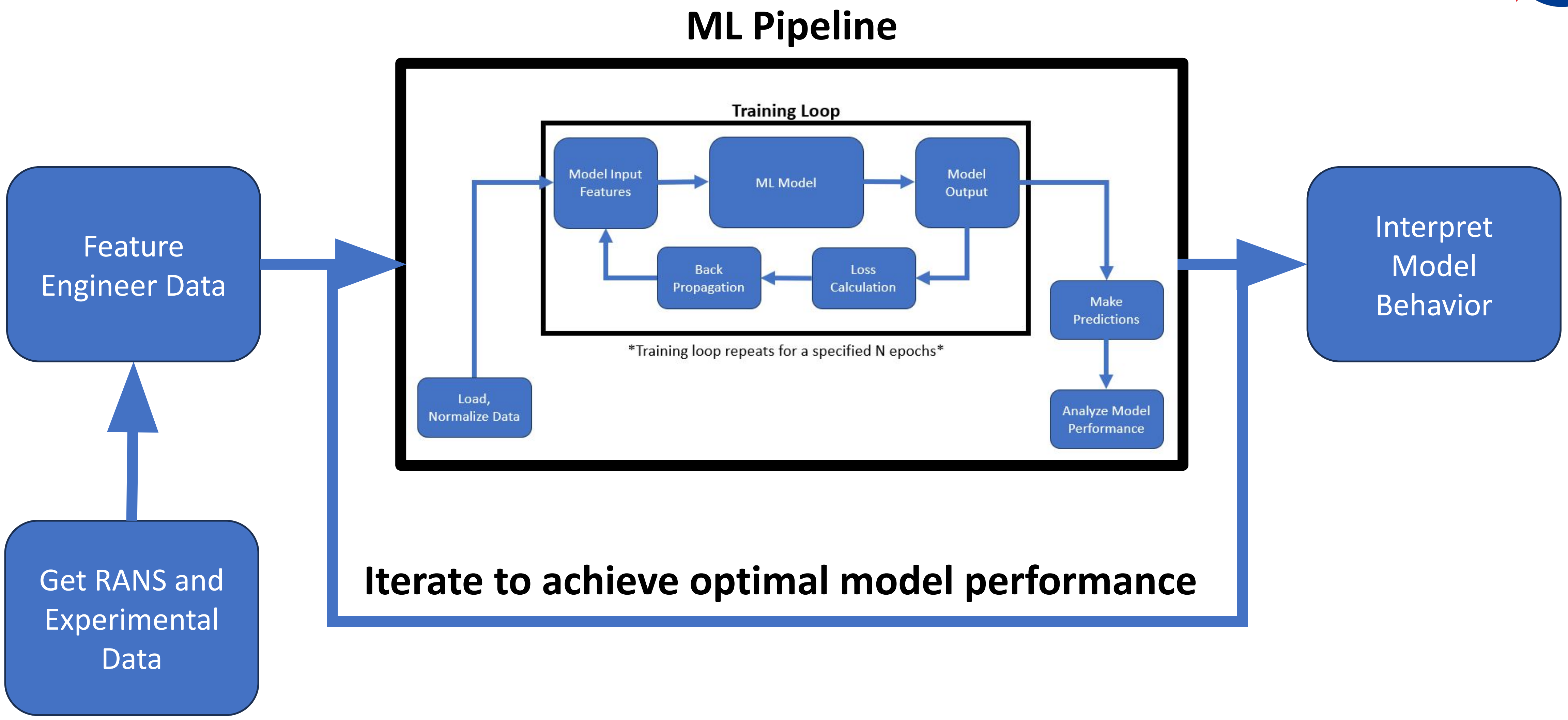
α_i vector can be dynamic over training. Ideally most importance on L_{Data}

Template

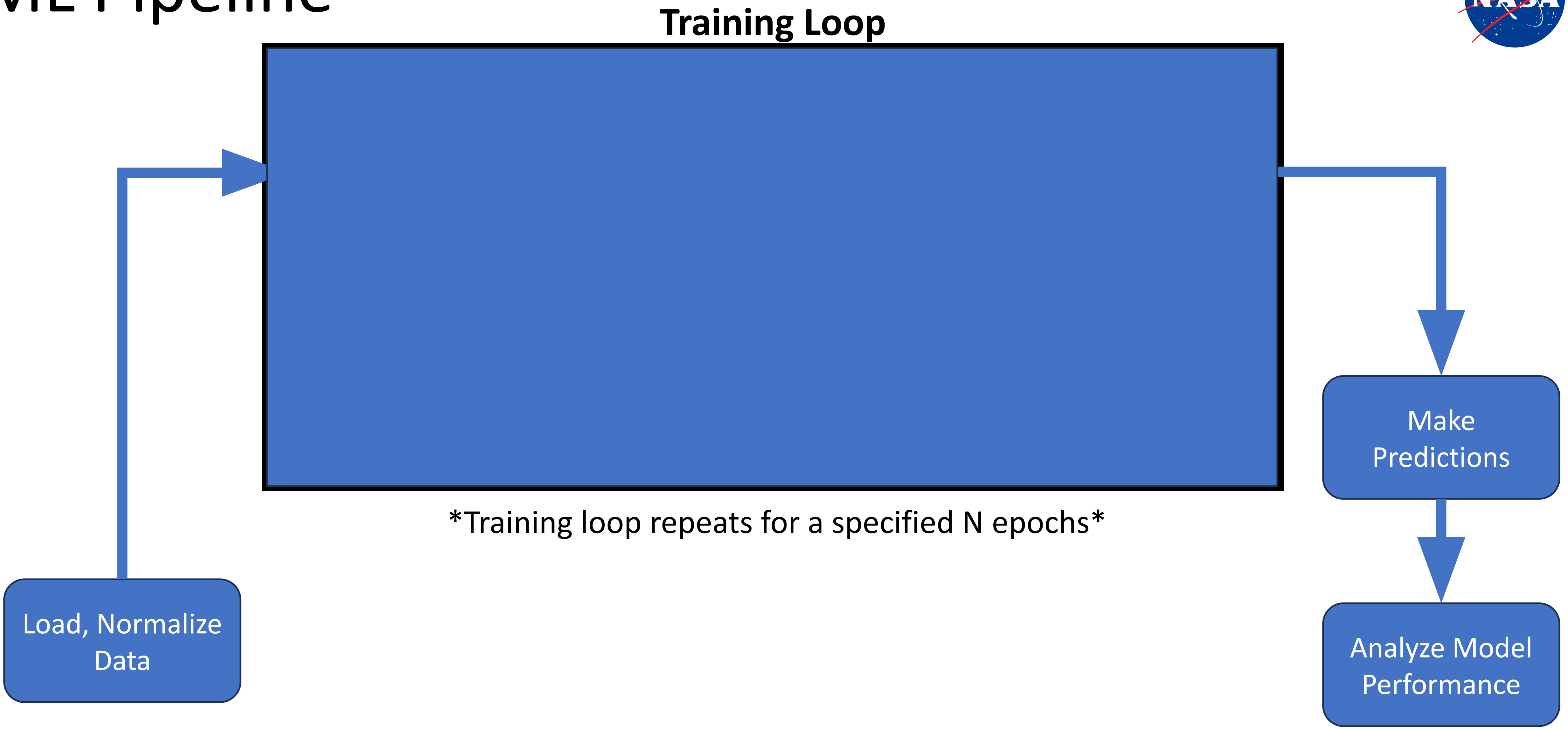
Project Workflow



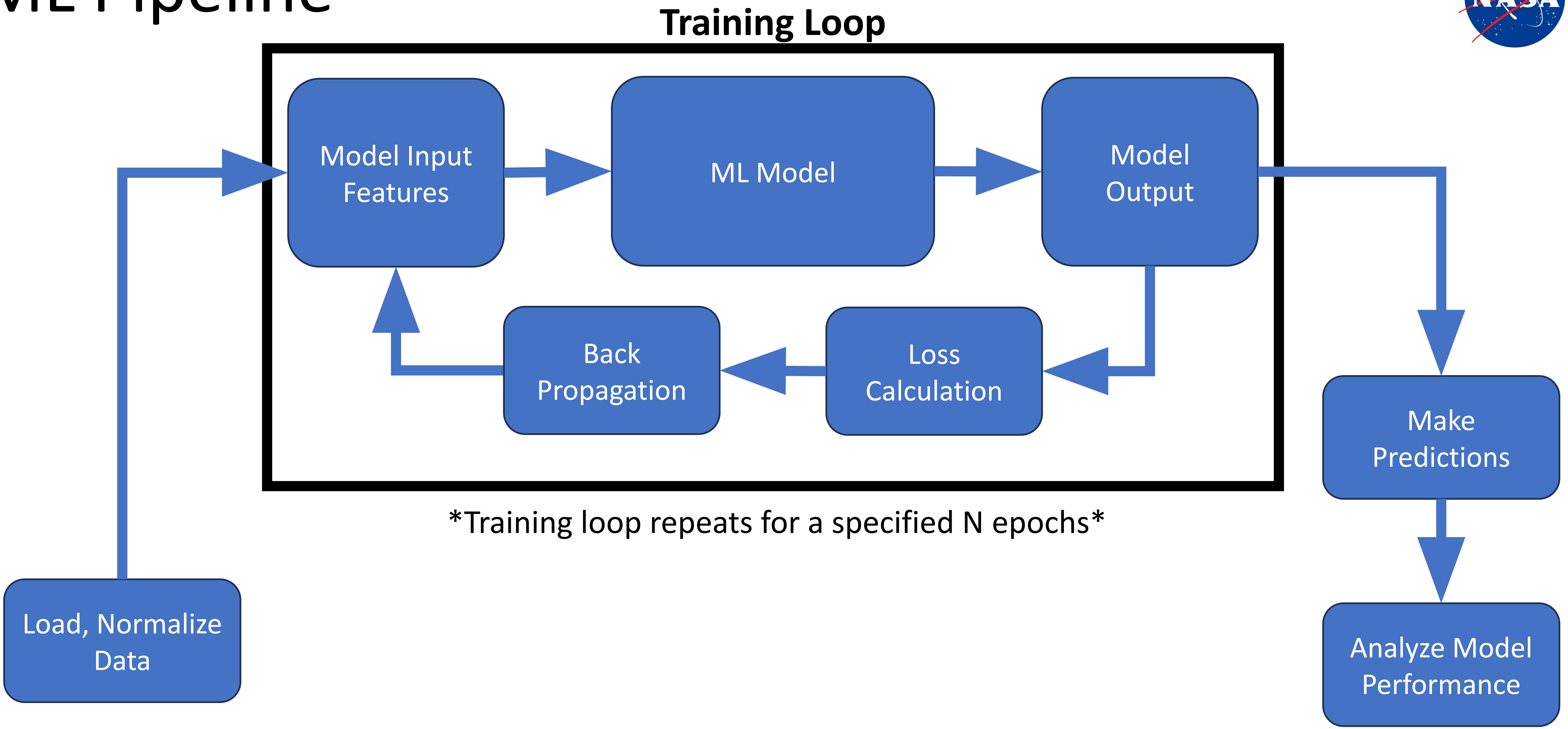
Project Workflow



ML Pipeline

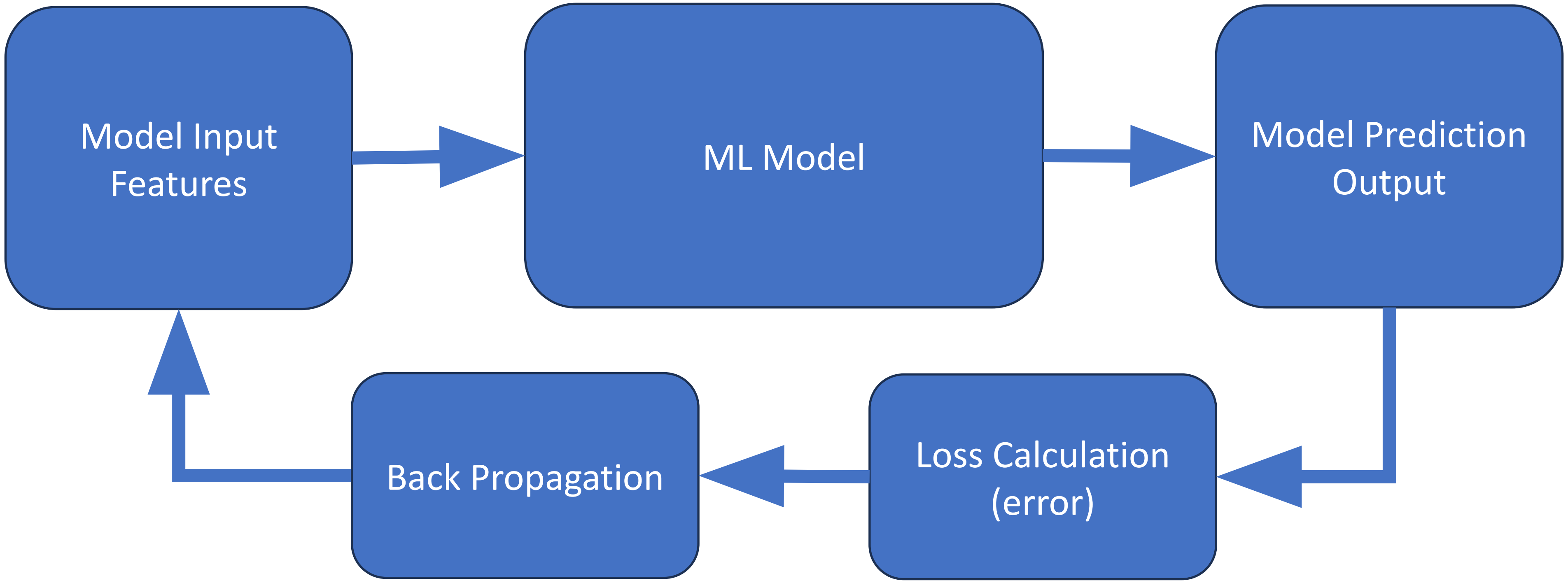

Training Loop
Training loop repeats for a specified N epochs
Load, Normalize Data
Make Predictions
Analyze Model Performance

ML Pipeline



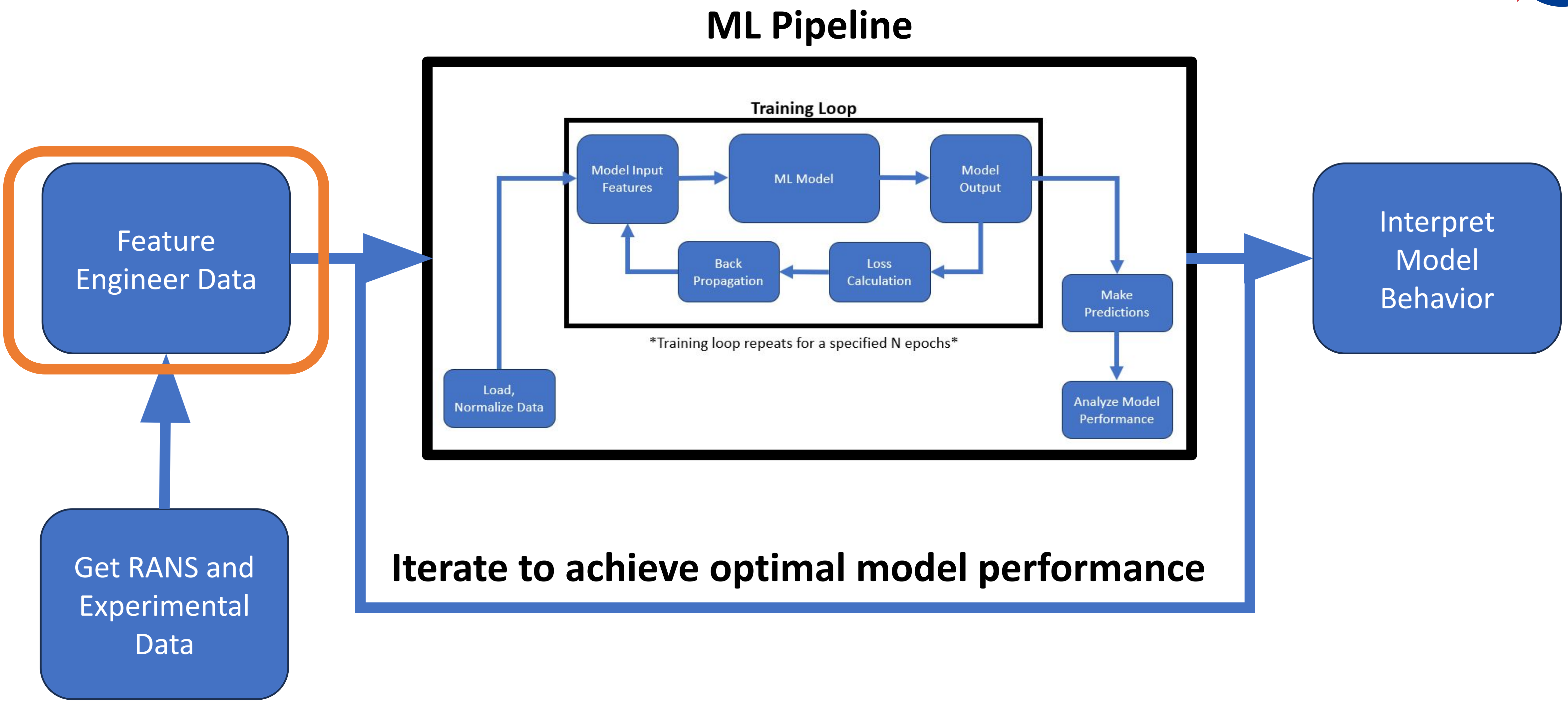
Training loop repeats for a specified N epochs

Training Loop

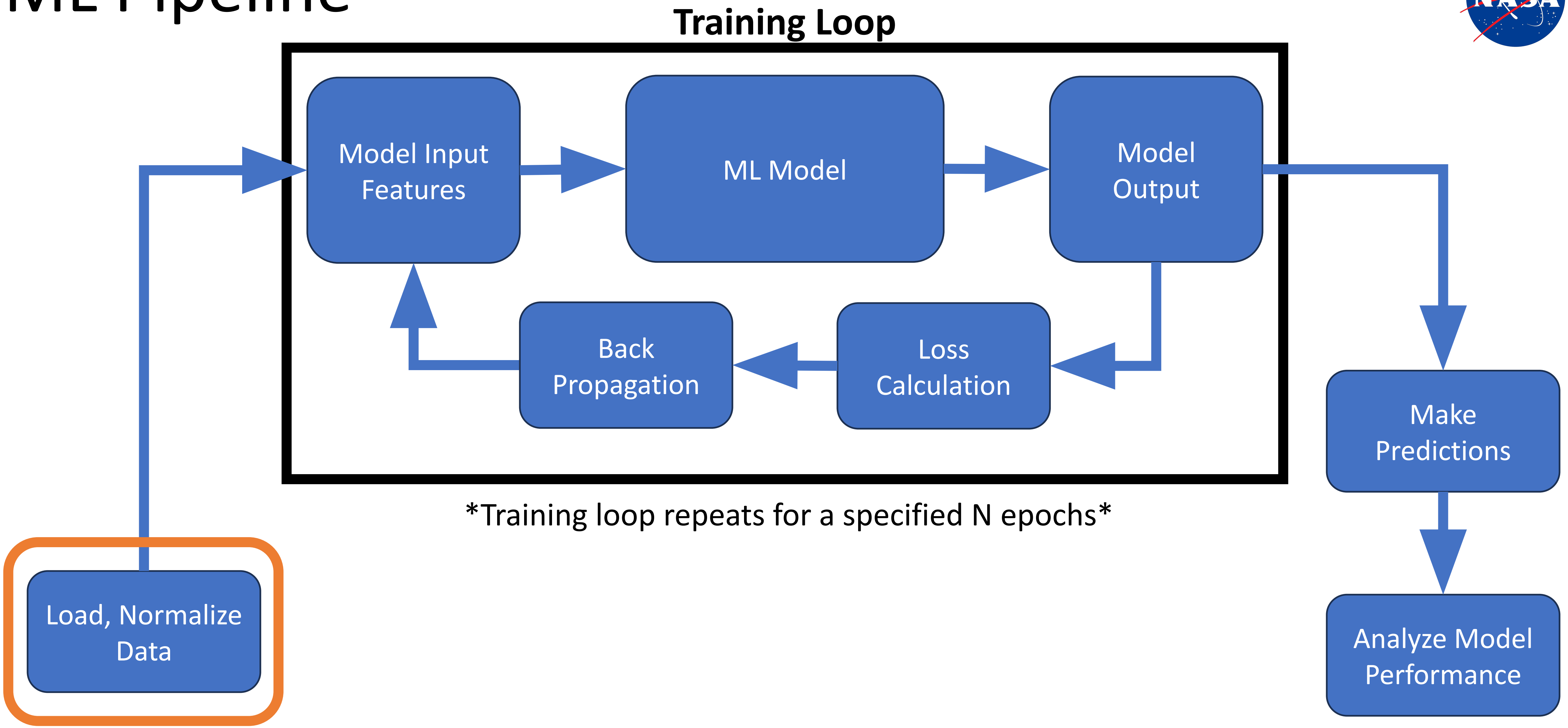


Training loop repeats for a specified N epochs

Project Workflow

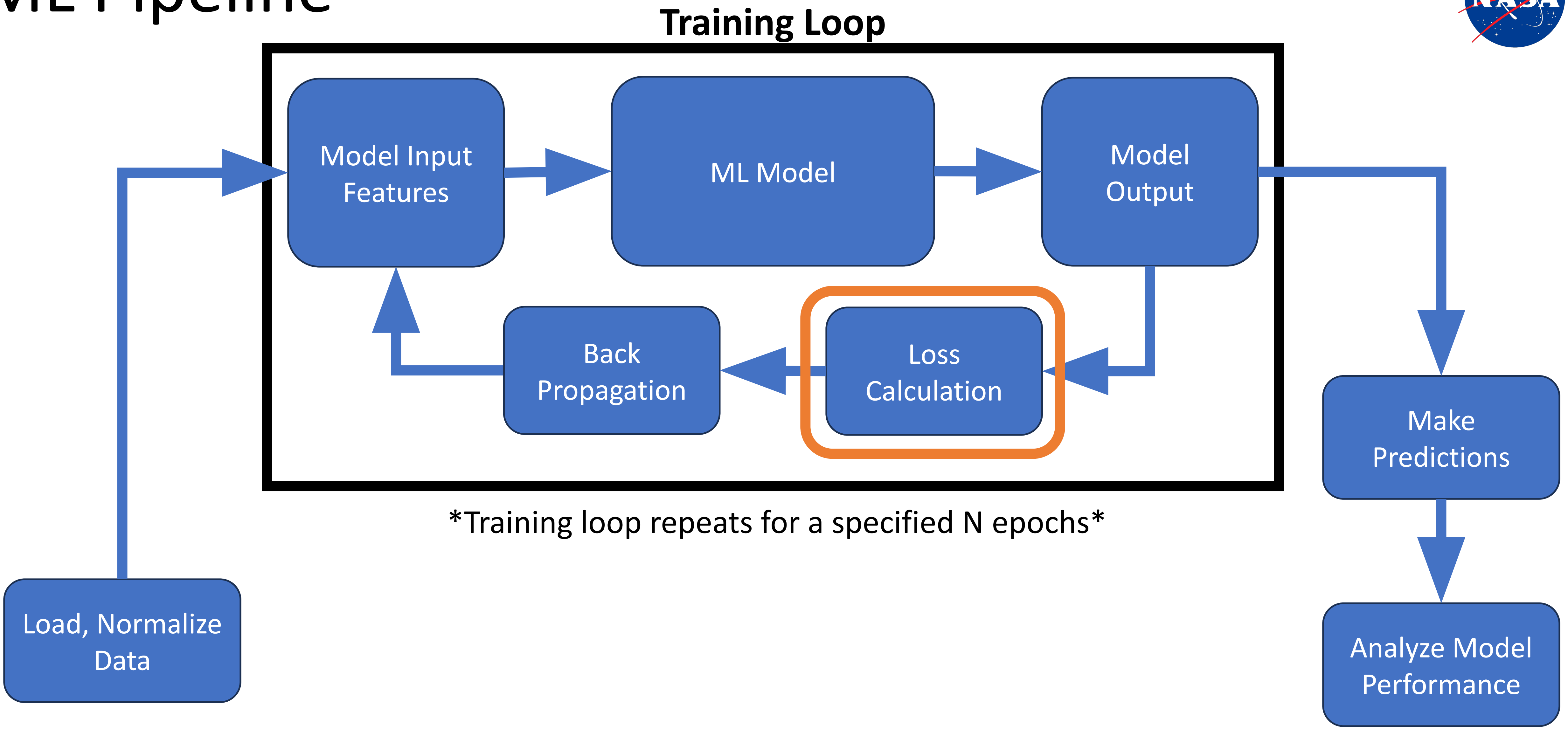


ML Pipeline



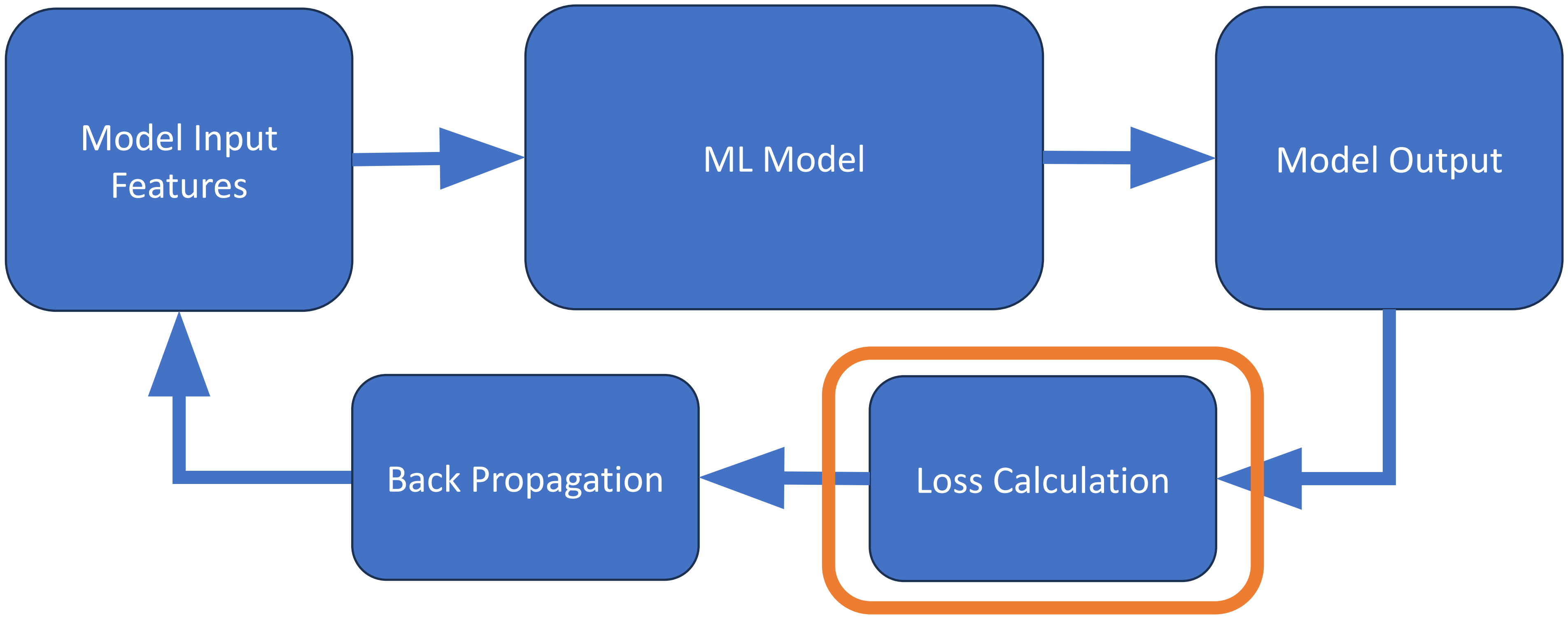
Training Loop
Model Input Features
ML Model
Model Output
Back Propagation
Loss Calculation
Training loop repeats for a specified N epochs
Load, Normalize Data
Make Predictions
Analyze Model Performance

ML Pipeline



Training loop repeats for a specified N epochs

Training Loop



Training loop repeats for a specified N epochs